

$$= 7.27 \pm 1.645 \times 0.69$$

$$= 7.27 \pm 1.135$$

$$\boxed{C.I = (6.135, 8.405)}$$

Conclusion: The mean time spent by students is b/w 6.135 hr & 8.405 hr //

- * In C.I, we are estimating the populⁿ parameter based on sample.
- * Margin of error decides L.B. cutoff & U.B cutoff
- * If there is more gap in C.I, there is less precision
- * So, we need to increase sample size, to get more precision.

* Hypothesis Testing:

- * Assumption abt the population parameters.
- * There are some claims.
- * H: Testing is statistical process to make the inferences abt the populⁿ parameters.
- * There are 2 claims & we draw conclusion from these claims.
- * If 2 claims are equal $\Rightarrow$ null hypothesis (H_0)
- * If 2 claims are diff $\Rightarrow$ Alternative hypothesis (H_1)/(H_A)
- H_0 : (Null Hypothesis): There is no diff or no relⁿ / no effect b/w 2 populⁿ (mean) parameters.
- H_1 / H_A : (Alternative Hypothesis): There is a diff / relⁿ / effect b/w 2 population parameters.
- * Null & Alternative hypothesis both are complementary statements.

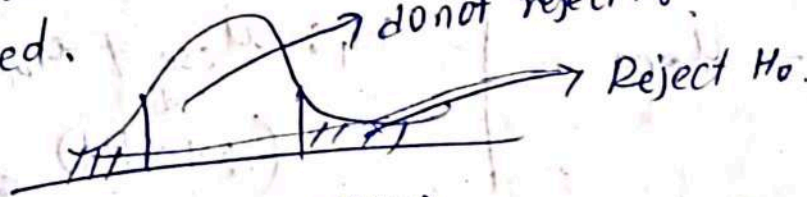
* The conclusions are drawn in 2 ways:-

1. Reject H_0 .

2. Do not reject H_0 $\rightarrow H_0$.

* If we want to say accept $H_0 \Rightarrow$ we need to test the entire populⁿ, but it is difficult & we are testing on sample, so we can say "do not reject H_0 " instead of "Accept H_0 ".

* we find ≥ 1 statistic values, & if it is in boundari, then H_0 is not rejected o/w H_0 is rejected.



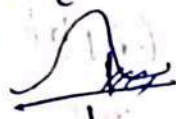
* There are 2 ways:-

1. One-tailed Test

$\downarrow$
rejectⁿ region will be at one side (left/right)



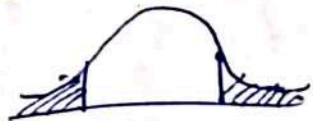
$\downarrow$
left-tailed



$\downarrow$
right-tailed.

2. Two-tailed Test.

$\Rightarrow$ rejectⁿ region will be at both sides.



statements:-

1. A student believes she scores less than the avg student who gets 75 marks.

$H_0: \text{Marks} \geq 75$

$\mu \geq 75$

$H_1: \text{Marks} < 75$

$H_1: \mu < 75$

(left tail)



2. You want to test if boys & girls have diff avg study times.

$H_0: \mu_b = \mu_g$

$H_1: \mu_b \neq \mu_g$

(2 tail)



* ' $\neq$ ' is not exist in H_2 .

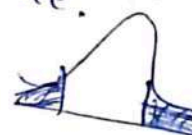
③ A researcher claims that Indian men are on avg more than 10 kgs heavier than India women. women avg is 65 kgs. What is null & Alternate hypothesis.

$$H_0: \mu_m \leq \mu_w \quad \left| \quad H_1: \overset{(75)}{\mu_m} > \overset{(65)}{\mu_w} \right.$$

(right tail) 

④ The avg salary of CSE & ECE students is diff. frame H_0 & H_1 .

$$H_0: \mu_{cse} = \mu_{ece} \quad \left| \quad H_1: \mu_{cse} \neq \mu_{ece} \right.$$

(2 tailed) 

⑤ A deodorant brand claims their deodorant fragrance stays more than 24 hrs. H_0 & $H_1 = ?$

$$H_0: \mu \leq 24 \text{ hrs} \quad \left| \quad H_1: \mu > 24 \text{ hrs} \right.$$

(right tail) 

* This hypothesis testing is done in 2 ways.

confidence Interval
Significant Test.

* Significant Test (steps)

1. Define H_0 & H_1 (null & Alternative hypothesis)
2. select significance level (α)
3. choose the appropriate Test $\left(\begin{array}{l} \text{Z-Test} \\ \text{T-Test} \\ \text{chi-square Test} \end{array} \right)$ based on the problem.
4. calculate the Test statistic
5. compare statistic value with critical value / p value.

6. Draw the conclusions.

1. If $p \leq \alpha \Rightarrow \text{Reject } H_0$.
2. If $p > \alpha \Rightarrow \text{Do not reject } H_0$.

Confidence Interval for population proportion (p)

$$* C.I = \left[\bar{p} \pm Z_{\alpha/2} \left(\sqrt{\frac{\bar{p} \cdot \bar{q}}{n}} \right) \right]$$

$\bar{p}$ = sample proportion, p = populⁿ proportion

$$\bar{q} = 1 - \bar{p}$$

$$C.I = \left[\underbrace{\bar{p} - Z_{\alpha/2} \sqrt{\frac{\bar{p} \cdot \bar{q}}{n}}}_{lb} < p < \underbrace{\bar{p} + Z_{\alpha/2} \sqrt{\frac{\bar{p} \cdot \bar{q}}{n}}}_{ub} \right]$$

① A retail store was interested in finding the proportion of customers ~~through~~^{who} pay-through cash for the merchandise. They buy at the store. For a sample of 100 customers, it was found that 70 customers paid by cash. calculate the 95% C.I for proportion of customers who paid by cash.

$$\Rightarrow \bar{p} = \frac{70}{100} = 0.7$$

$$\bar{q} = 1 - 0.7 = 0.3$$

$$n = 100, Z_{\alpha/2} = 1.96$$

$$\therefore C.I = \bar{p} \pm Z_{\alpha/2} \sqrt{\frac{\bar{p} \bar{q}}{n}}$$

$$= 0.7 \pm 1.96 \sqrt{\frac{0.7 \times 0.3}{100}}$$

$$= 0.7 \pm 1.96(0.045)$$

$$= 0.7 \pm 0.0882$$

$$C.I = [0.6118, 0.7882]$$

$$pe [0.61 < p < 0.789]$$

$\therefore$ we are 95% confident that the interval is 0.61, 0.789 contains the true populⁿ proportion of the customers who pay-by-cash.

* Hypothesis Testing

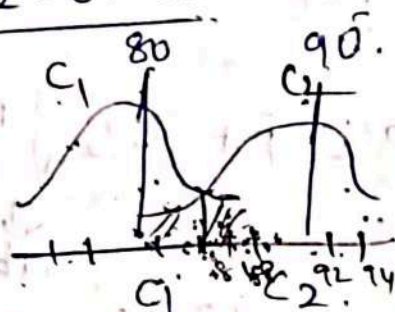
* Hypothesis $\rightarrow$ assumption / claim / belief.

* It is done in 2 ways.

1. one tailed $\rightarrow$ rejection region is at right or left
 (left tailed / right tailed)

2. two-tailed $\rightarrow$ Rejection region is on both sides

Type-I-error



Type I - reject H_0

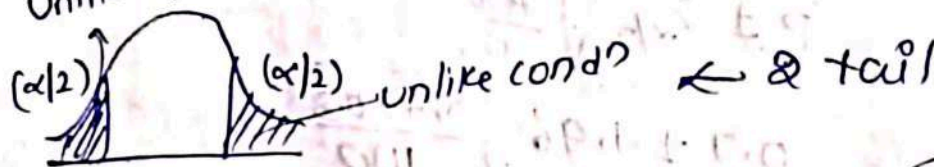
Type II - failed to reject H_0

C1 class avg = 80, C2 class avg = 90
 * we have a student with 82 marks in C2, but 82 is near to 80, so we believe that 82 mark student is from C1. This is Type-I error.

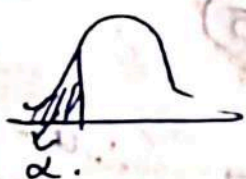
* conditional prob of rejecting a null hypothesis when it is true is called Type-I error (or) false positive error.

* Type-I error is indicated with α .

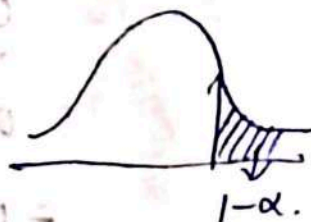
* so we call ' α ' as significance value or unlikely condⁿ.



left tail $\rightarrow$



Right tail



* statements

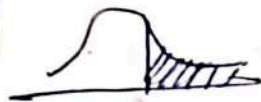
① Avg annual salary of machine learning experts is diff for males & females. frame H_0 & H_1

$$H_0: \mu_m = \mu_f \quad | \quad H_1: \mu_m \neq \mu_f$$

② on an avg people with phd in analytics earn more than people with phd in engineering.

$$H_0: \mu_A \leq \mu_E \quad | \quad H_1: \mu_A > \mu_E$$

→ right tailed



$H_1: \mu_m > \mu_f \Rightarrow$ right tailed $H_2: \mu_m < \mu_f \Rightarrow$ left tailed $H_1: \mu_m \neq \mu_f \Rightarrow$ 2 tailed.	**
--	----

* Type II - Error

* conditional prob of failing to reject a null hypothesis when the alternative hypothesis is true is called "Type-II error (or) false negative error."

* It is indicated with β .

Statements

① Avg life of vegetarians is more than meat eaters. frame H_0 & H_1

$$H_0: \mu_v \leq \mu_n \quad | \quad H_1: \mu_v > \mu_n$$

→ right tailed →



② The salary of ML experts on avg is at least 1 lakh dollars. frame H_0 & H_1

$$H_0: \mu \leq 1 \text{ lakh} \quad | \quad H_1: \mu > 1 \text{ lakh}$$

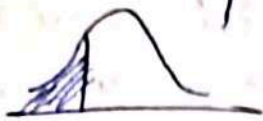
right tailed →



③ Avg waiting time at the london hitoo security check is less than 30 min. frame H_0 & H_1 .

$$H_0: \mu \geq 30 \text{ min}$$

$$H_1: \mu < 30 \text{ min}$$

left tail $\rightarrow$ 

④ Avg annual salaries of male & female MBA students are diff at the time of graduation.
frame H_0 & H_1 .

$$H_0: \mu_m = \mu_f$$

$$H_1: \mu_m \neq \mu_f$$

$\rightarrow$ 2 tailed



α value	left tailed	Right tailed	2-tailed
(90%) 0.1	-1.28	+1.28	-1.64 to +1.6
(95%) 0.05	-1.64	+1.64	-1.96 to +1.96
(99%) 0.01	-2.33	+2.33	-2.58 to +2.5
(96%) 0.04	-1.75	+1.75	-2.05 to +2.0

* Hypothesis Tests

- ① Z-Test $\rightarrow$ ^{known} populⁿ std deviation, $n \geq 30$.
- ② T-Test $\rightarrow$ unknown ~~populⁿ~~ std deviation of populⁿ σ & $n < 30$.
- ③ chi-square Test $\rightarrow$ data is categorical form.
- ④ F-Test $\rightarrow$ variance.

* Z-Test

* one sample z-test $\rightarrow$ only one mean, sample size one.

* Two sample z-test $\rightarrow$ 2 means, & sample size

* T-Test

- ① one sample.
- ② 2 sample T-test $\leftarrow$ with equal variance ($\sigma_1 = \sigma_2$)
with unequal variance ($\sigma_1 \neq \sigma_2$)
- ③ paired t-test

③ chi-squared: $\rightarrow$ 2. Independence.
* either of ~~good~~ ^{fit} test, ~~features of~~ ~~fit~~
④ F-Test: $\rightarrow$ variance ANOVA.

2. Independence.

③ chi-square test, ~~either of~~ ^{fit} ~~variance~~ ANOVA.

* F-Test: - variance ANOVA.

① An agency base out of Bangalore claimed that the avg monthly disposable income of families living in Bangalore is greater than (INR) 4200 with a std deviation (INR) 3200. From a random sample of 40,000 families, The avg disposal income was estimated as (INR) 4250. Assume that the popl std devⁿ $\sigma = 3200$. conduct an appropriate hypothesis test at 95% confidence level to check the validity of the claim by the agency?

sol: 7 1) frame, H_0 & H_1 .

$$\sigma = 3200. \quad \mu = 4200.$$

$$n = 40100 \quad \bar{x} = 4250.$$

i) income of families $H_0: \mu \leq 4200$

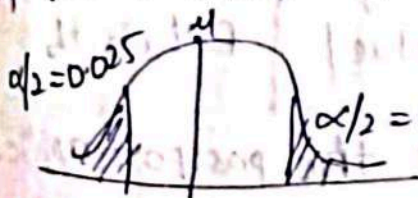
$$H_1: \mu > 4200 \rightarrow \text{right tail}$$

2) decide α value is 95% $\Rightarrow \alpha = 0.05$

3) choose appropriate Test-statistic

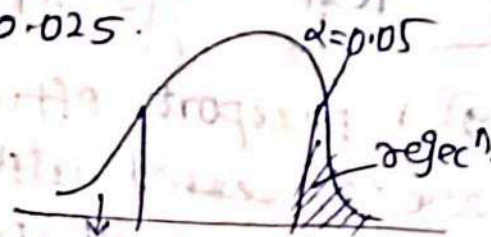
$$Z = \frac{\bar{X} - \mu}{\left(\frac{\sigma}{\sqrt{n}} \right)} = \frac{4250 - 4200}{\left(\frac{3200}{\sqrt{40000}} \right)} = \frac{50}{\left(\frac{3200}{200} \right)} = \frac{50}{16} = 3.125$$

4) Compare statistic value with critical value (or) p value with α .



$\alpha = 0.05$

$$\alpha/2 = 0.025$$



* total region will be right side

$$3.125 > 1.64$$

fail to
reject.
do not reject

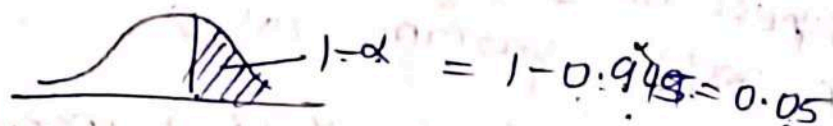
z-table value (1.64)

$$= 0.94\$$$

* In left tail.



* In right tail



* In 2 tail.



* Here, in z statistical value $> c.v$

$$[3.125 > 1.64]$$

$$p(3.125) = 0.99910 \text{ (for right side } 1-\alpha)$$

$$1 - 0.99910 = 0.0009$$

$$prob \Rightarrow 0.0009 < 0.05 \Rightarrow \boxed{p < \alpha}$$

⑤ conclusion based on condⁿ:-

$\boxed{p < \alpha, \text{ so reject of null hypothesis } (H_0)}$

* condition for rejection of Null hypothesis H_0

Type of Test	cond ⁿ	Decision
Left-tailed Test	$z\text{-st} < \text{critical value}$	Reject H_0
	$z\text{-st} \geq \text{critical value}$	Retain H_0
Right-tailed Test	$z\text{-st} > \text{critical value}$	Reject H_0
	$z\text{-st} \leq \text{critical value}$	Retain H_0
Two-tailed Test	$ z\text{-st} > \text{critical value} $	Reject H_0
	$ z\text{-st} \leq \text{critical value} $	Retain H_0

② A passport office claims that the passport applicants are processed within 30 days of submitting the applicⁿ and all necessary documents. processing time of 40 passport applicants is given in table below the $\downarrow$ standard deviation of the Populⁿ

processing time is 12.5 days. Conduct a hypothesis test at significance level $\alpha = 0.05$ to verify the claim made by the office passport processing time. $\rightarrow 16, 16, 30, 37, 25, 22, 19, 35, 27, 32, 34, 28, 24, 35, 24, 21, 32, 29, 24, 35, 28, 29, 18, 31, 28, 33, 32, 24, 25, 22, 21, 27, 41, 23, 23, 60, 29, 38, 26, 28.$

Sol. $\sigma = 12.5, \alpha = 0.05, \bar{X} = 28.15, n = 40.$

$$\bar{X} = \frac{16 + 16 + \dots + 28}{40} = 28.15$$

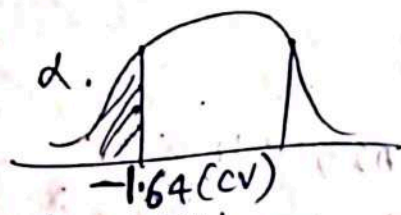
i) $H_0: \mu \geq 30 \text{ days} \mid H_1: \mu < 30 \text{ days}$ (left tail)

ii) Decide α value $= \alpha = 0.05$.

iii) chose appropriate Test-statistic

$$Z = \frac{\bar{X} - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)} = \frac{28.15 - 30}{\left(\frac{12.5}{\sqrt{40}}\right)} = \frac{-1.85}{1.976} = -0.936$$

iv) compare statistic value with st. value.



Z-st. CV

$$-0.936 > -1.64$$

$$P(-0.936) = 0.17361 \quad (\text{for left tail} \Rightarrow \alpha)$$

$$\therefore 0.17361 > 0.05 \quad (P > \alpha)$$

v) conclusion: $P > \alpha \therefore$ so, failed to reject null hypothesis (H_0)

* popular Hypothesis Tests

- 1) Z-Test 2) T-Test 3) chi-square Test 4) F-Test

① Z-Test $\rightarrow$ $\begin{cases} \text{one sample Z-Test} \\ \text{Two sample Z-Test} \end{cases}$

② T-Test $\rightarrow$ $\begin{cases} \text{one sample T-Test} \\ \text{Two " " with equal variances} \\ \text{Two " " with unequal variances} \\ \text{paired T-Test} \end{cases}$

③ chi-square Test $\rightarrow$ fitness of good Test

④ F-Test $\rightarrow$ Anova Test

Requirements / Assumptions

① Z-Test: $n \geq 30$ & known popⁿ std devⁿ (σ)

② T-Test: $n < 30$ & unknown popⁿ std devⁿ (σ)

③ chi-square based on categorical data

④ F-Test based on frequency.

* If α is not given $\rightarrow$ consider confidence level = 95%
 $\alpha = 0.05$ (for Z-Test).

* one-sample Z-Test problems

* for 2 sample, they give $n_1, n_2, \mu_1, \sigma_2$.

* for 1 " they give n, σ . Form: $Z_{st} = \frac{\bar{x} - \mu}{(\sigma/\sqrt{n})}$

* for 1 " they give n, σ .
 ① A manufacturer produces bolts with a thickness of exactly 1 inch. A customer takes a random sample of 100 bolts & found that $\bar{x} = 1.2$ inches & $s = 0.40$ inches. should the manufacturers claim that the bolts are exactly 1 inch be rejected. Test at $\alpha = 0.01$

sol: 1) Define H_0 & H_1

$H_0: \mu = 1 \text{ inch}$ | $H_1: \mu \neq 1 \text{ inch}$
 (2-tailed test)



2. select significance level (α)

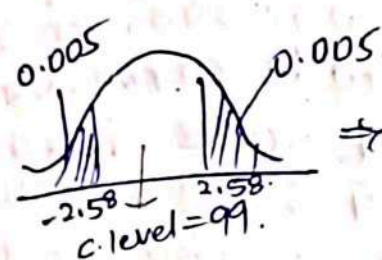
$$\alpha = 0.01$$

3. choose appropriate test statistic ($n \geq 30 \Rightarrow Z$ -Test)

$$Z_{st} = \frac{\bar{x} - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)} \quad \text{or} \quad \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}}\right)} \quad (\text{sample std dev})$$

$$= \frac{1.2 - 1}{\left(\frac{0.4}{\sqrt{100}}\right)} = \frac{0.2 \times 10}{0.4} = \frac{2}{0.4}$$

$$Z_{st} = 5$$



$\Rightarrow$ Rejection at both regions.
 $\alpha/2 = \frac{0.01}{2} = 0.005$

* Find the value at with the 0.005 lies.
 critical. c.v | Prob.

i.e. (-2.58 to +2.58)

(5 > 2.58) $\Rightarrow$ extreme right. (in rejection region)
 $\downarrow$
 z-st c.v

5. Comp p-value with α .

~~For z-table~~
 $P(5) = 0$

(For z-table.
 i.e. $P(> 3.9) = 0$
 $P(< -3.9) = 0$)

$$P(5) = 1 - 0.9999 \approx 0 \quad (\text{b/c it is at right}) \rightarrow (\text{no need})$$

$P < \alpha$
 $0 < 0.0099$

$\therefore P < \alpha \Rightarrow$ Reject H_0 .

6. conclusion of

$P < \alpha$, rejection of null hypothesis (H_0)

i.e. we have enough evidence to reject null hypothesis

$$C.I = \bar{x} \pm Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}}\right)$$

$$Z_{\alpha/2} = 2.576$$

$$= 1.2 \pm 2.576 \left(\frac{0.4}{\sqrt{100}}\right) = 1.2 \pm 2.576 (0.04) = 1.2 \pm 0.10304$$

$$C.I = [1.096, 1.303] \quad (\mu=1) \rightarrow \text{popl}^n \text{ mean}$$

If μ lies in boundaries $\Rightarrow$ do not reject H_0
 If μ lies out of boundaries $\Rightarrow$ reject H_0

Here μ lies in rejection region. (out of boundaries)
 $\therefore$ we reject null hypothesis (H_0). //

② A company claims that a box of ~~raisins~~ bran cereals contains at least 100 raisins. An investigator working for the federal trade commission takes a random sample of 324 boxes of cereal and finds that $\bar{x} = 97$ raisins & $s = 9$ raisins. Should the company's claim be rejected. Test at $\alpha = 0.05$.

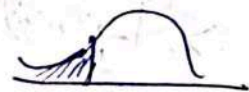
① Define H_0 & H_1 .

$$H_0 = \mu \geq 100$$

(at least $\rightarrow$ greater than or equal)

$$H_1: \mu < 100$$

(left tail)

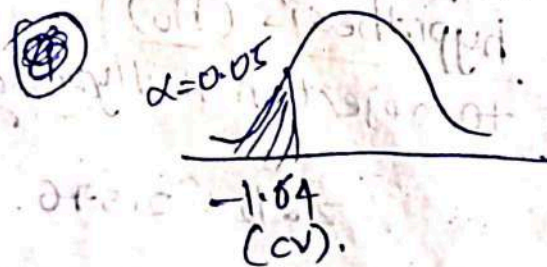


$$\bar{x} = 97, \mu = 100, n = 324, s = 9.$$

② select significance level (α)
 $\alpha = 0.05$

③ choose appropriate test statistic. ($n \geq 30 \Rightarrow z$ -test)

$$z = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}}\right)} = \frac{97 - 100}{\left(\frac{9}{\sqrt{324}}\right)} = \frac{-3}{\left(\frac{9}{18}\right)} = -6$$



$$-6 < -1.64$$

(z-st) (C.V.)
 (-6 falls on extreme left $\rightarrow$ rejection region)

④ compare P & α value

$$P(-6) = 0 \quad \& \quad \alpha = 0.05$$

$$0 < 0.05 \Rightarrow P < \alpha$$

5) Conclusion:

$P < \alpha \Rightarrow$ Reject Null hypothesis (H_0)

i.e. we have enough evidence to reject Null hypothesis.

$$\begin{aligned} \text{C.I.} &= \bar{x} \pm Z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right) \\ &= 97 \pm 1.96 \left(\frac{9}{\sqrt{324}} \right) \\ &= 97 \pm 1.96 (0.5) \\ &= 97 \pm 0.98 \end{aligned}$$

$$Z_{\alpha/2} = 1.96$$



$$\text{C.I.} = [96.02, 97.98]$$

$$\mu = 100$$

$$\text{C.I.} = [96.02, 97.98]$$

i.e. Here μ lies out of boundaries (Rejection region)

$\therefore$ we reject Null hypothesis (H_0)

3) A researcher claims that 10 yr olds watch 6.6 hrs of TV daily. You try to verify this with the following sample data $n=100$, $\bar{x}=6.1$ hrs, $s=2.5$ hrs, Test at $\alpha=0.01$.

Sol: ① Define H_0 & H_1

$$H_0: \mu = 6.6 \text{ hrs}$$

$$H_1: \mu \neq 6.6 \text{ hrs.}$$

(2-tail test)

② Select significance level (α)

$$\alpha = 0.01$$

③ Choose appropriate test statistic ($n \geq 30 \Rightarrow$ z-test)

$$Z = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}} \right)} = \frac{6.1 - 6.6}{\left(\frac{2.5}{\sqrt{100}} \right)} = \frac{-0.5 \times 10}{2.5} = \frac{-5}{2.5} = -2$$



$\Rightarrow$ 2 tail. ($\frac{\alpha}{2} = 0.005$)

$$\Rightarrow Z\text{-st} = |2| < |2.58|$$

Res at $\alpha = 0.01$ (do not reject H_0)

4) Compare p & α value.

$$p(-2) = 0.02275$$

$$\alpha = 0.01$$

$$\Rightarrow \boxed{p > \alpha}$$

5) Conclusion

$p > \alpha \Rightarrow$ Do not reject null hypothesis (H_0)

$$C.I = \bar{X} \pm z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right) \quad (2\alpha/2 = 2.576)$$

$$= 6.1 \pm 2.576 \left(\frac{2.5}{\sqrt{100}} \right)$$

$$= 6.1 \pm 2.576 \times 0.25$$

$$= 6.1 \pm 0.644$$

$$\boxed{C.I = [5.456, 6.744]} \quad (\mu = 6.6)$$

* Here μ lies in boundaries

$\therefore$ we do not reject H_0

4) A wine manufacturer claims that his wine has at most 9 ppm impurities in a barrel of wine.

$n=64$, $\bar{X}=9.06$ ppm impurities, $s=0.12$ ppm. Test

at $\alpha = 0.01$.

sol. 1) Define H_0 & H_1

$H_0: \mu \leq 9$ | $H_0 = \mu > 9$ (right tail)

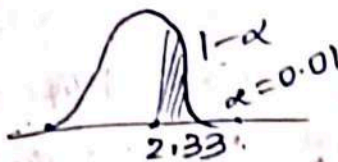
2) select the significant level (α)

$$\boxed{\alpha = 0.01}$$

3) choose appropriate Test statistic ($n > 30$, z -test)

$$Z_{st} = \frac{\bar{X} - \mu}{\left(\frac{s}{\sqrt{n}} \right)} = \frac{9.06 - 9}{\left(\frac{0.12}{\sqrt{64}} \right)} = \frac{0.06}{0.12} \times 8$$

$$\boxed{Z_{st} = 4}$$



$$Z_{st} = 4$$

$$(\alpha = 0.01)$$

$Z_{st} > C.V. \rightarrow$ lies in the reject region.

④ compare p & α value.

$$p(4) = 0 \text{ \& } \alpha = 0.01$$

$$(p(4) = 0 \Rightarrow \text{right tail})$$

$$\therefore 0 < 0.01 \Rightarrow \boxed{p < \alpha}$$

⑤ Conclusion:-

$p < \alpha \Rightarrow$ Reject null hypothesis (H_0).

i.e. we have enough evidence to reject null hypothesis

$$C.I = \bar{X} \pm Z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right)$$

$$Z_{\alpha/2} = 2.576$$

$$= 9.06 \pm 2.576 \left(\frac{0.12}{\sqrt{64}} \right)$$

$$= 9.06 \pm 0.03864$$

$$\boxed{C.I = [9.02, 9.098]}$$

$$\boxed{\mu = 9}$$

$\therefore$ Here ' μ ' lies in/out of boundaries (reject region)
 $\therefore$ we reject null hypothesis (H_0) //

⑤ The Just like Evampure water company claims that there are at most 1 ppm units of urine in its highly overpriced water. Use the following sample data test at $\alpha = 0.05$, $n = 121$ bottles,

$$\bar{x} = 1.1 \text{ ppm}, s = 0.33 \text{ ppm}$$

Q14 ① Define H_0 & H_1

$$H_0: \mu \leq 1 \text{ ppm}$$

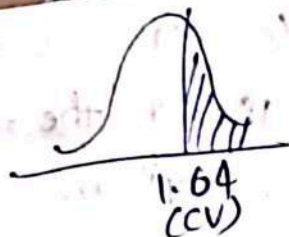
$$H_1: \mu > 1 \text{ ppm (right tail)}$$

② Select the significant level (α)

$$\boxed{\alpha = 0.05}$$

③ choose appropriate Test statistic ($n > 30$, 2-Test)

$$Z_{st} = \frac{\bar{X} - \mu}{\left(\frac{s}{\sqrt{n}} \right)} = \frac{1.1 - 1}{\left(\frac{0.33}{\sqrt{121}} \right)} = \frac{0.1}{0.33} \times 11 = \boxed{3.33}$$



$$\alpha = 0.05$$

$$Z_{st} = 3.33 > C.V = 1.64$$

$[Z_{st} > CV] \Rightarrow$ lies in rejection region.

④ compare p & α value

$$p(3.33) = 0.99952. \text{ (right tail)}$$

$$= 1 - 0.99952 = 0.00048.$$

$$\therefore [0.00048 < 0.05] : (p < \alpha),$$

⑤ conclusion

$[p < \alpha] \Rightarrow$ we reject null hypothesis (H_0) //

$$C.I = \bar{x} \pm Z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right) = 1.1 \pm 1.96 \left(\frac{0.33}{\sqrt{121}} \right)$$

$$= 1.1 \pm 0.0588$$

$$[C.I = [1.0412, 1.1588]] \quad \mu = 1$$

Here μ lies out of boundaries.

$\therefore$ we reject null hypothesis (H_0) //

* Two-Sample Z-Test

* sample size > 30 & populⁿ std devⁿ should be known.

* Assumptions $\Rightarrow$

1. The sample sizes n_1 & n_2 of 2 samples drawn from 2 populations are large ($n_1 \geq 30$ & $n_2 \geq 30$) (atleast 30)
2. The samples are drawn from 2 normally distributed populations with corresponding std deviations σ_1 & σ_2 are known.

* Formula $\Rightarrow$

$$Z_{st} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

($\sigma = s$ if s is given & σ is not given)

① The dean of St. Peter school of Mgt education believes that the graduating students with specialization in marketing earn at least 5000 more per month than the students with specialization in operations mgt. To verify this belief, the dean collected a sample data from his graduating students, conduct an appropriate hypothesis test at $\alpha = 0.05$ to check whether the diff in monthly salary is at least 5000 more for students with marketing specialization compared to oper's mgt. specialization. Assume that the salary of students of marketing & oper's spec follows normal distrib. sample values on marketing & operations students

specializ ⁿ	sample size	estimated mean of salary	popul ⁿ std dev ⁿ
Mkt	120 (n_1)	67500 ($\bar{x}_1$)	7200 (σ_1)
oper's Mgt	45 (n_2)	58950 ($\bar{x}_2$)	4600 (σ_2)

solⁿ ① define H_0 & H_1

$$H_0 = \mu_1 - \mu_2 \leq 5000$$

$$H_1: \mu_1 - \mu_2 > 5000$$

(right tail)

② Select the significant level (α)

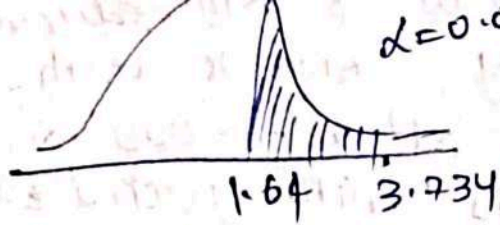
$$\alpha = 0.05$$

③ choose appropriate Test (2-sample z-test).
($\mu_1 - \mu_2 = 5000$)

$$z_{st} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$= \frac{(67500 - 58950) - (5000)}{\sqrt{\frac{7200^2}{120} + \frac{4600^2}{45}}}$$

$$= \frac{3550}{\sqrt{432000 + 470222.22}} = \boxed{3.737}$$



$z_{st} = 3.737 > c.v. (1.64)$
lies in reject region.

(4) compare p & α

$$p(3.737) = 0.99991. \quad (1 - 0.99991 = 0)$$

$$p \approx 0 \text{ \& } \alpha = 0.05$$

(5) Compare p & α .

$\therefore \boxed{p < \alpha} \Rightarrow$ we reject the null hypothesis (H_0)

we have enough evidence to reject H_0 .
i.e. H_1 is blm true $\Rightarrow$ there is a diff in salary of marketing & opes's mgt.

(2) Typing speed on a PC $\Rightarrow$ who types faster men or women

	men	women
$\bar{x}$	65 WPM	68 WPM
s	10 WPM	14 WPM
n	50	60

Test at $\alpha = 0.01$.

Soln: (1) Define H_0 & H_1

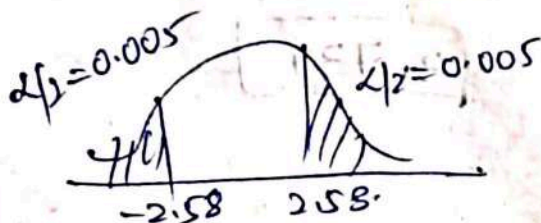
$$H_0: \mu_1 - \mu_2 = 0 \quad | \quad H_1: \mu_1 - \mu_2 \neq 0$$

(2) Select the significant level (α) (2 tail)

$$\boxed{\alpha = 0.01}$$

(3) choose appropriate Test

$$z_{st} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{10^2}{50} + \frac{14^2}{60}}} = \frac{(65 - 68) - (0)}{\sqrt{\frac{100}{50} + \frac{196}{60}}}$$



$$= \boxed{-1.307}$$

$$|z_{st}| = 1.307$$

$$|c.v| = 2.58$$

$|z_{st}| < c.v \Rightarrow$ lies in do not reject region.

④ compare p & α .

$$p(-1.307) = 0.08534, \quad \alpha = 0.01 \Rightarrow \boxed{p > \alpha}$$

⑤ conclusion:

$\boxed{p > \alpha} \Rightarrow$ we do not reject null hypothesis (H_0)
i.e. There is no diff b/w typing speed of men & women

③ Take home pay $\Rightarrow$ who earns more married or unmarried people.

	Married
$\bar{x}$	639.60 \$
s	60 \$
n	40

Unmarried

658.20 \$

90 \$

60

Test at $\alpha = 0.04$.

sol: ① Define H_0 & H_1 .

$$H_0: \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 - \mu_2 \neq 0 \text{ (2-tail)}$$

② select the significant level (α)

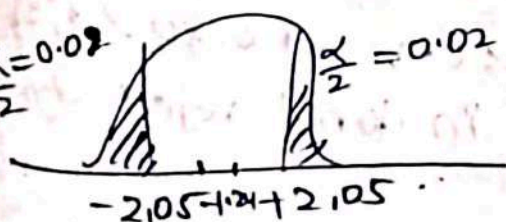
$$\boxed{\alpha = 0.04}$$

③ choose appropriate Test.

$$z_{st} = \frac{(639.60 - 658.20) - (0)}{\sqrt{\frac{(60)^2}{40} + \frac{(90)^2}{60}}} = \boxed{-1.24}$$

c.v for $\alpha = 0.04$

$\frac{\alpha}{2} = 0.02$



$$z_{st} = |-1.24| = 1.24 < 2.05$$

lies in do not reject region.

④ compare p & α

$$p(-1.24) = 0.10749 \text{ \& } \alpha = 0.04$$

$$\boxed{p > \alpha}$$

② conclusion

$[P > \alpha]$ we do not reject null hypothesis (H_0)

i.e. There is no diff b/w earning of married & unmarried people (both will each the same) //

④ Who is longer married or unmarried woman. Test at $\alpha = 0.01$.

Unmarried

$$\bar{x} = 78.54$$

$$s = 14.04$$

$$n = 140$$

married

$$77.04$$

$$16.04$$

$$160$$

Sol: Define H_0 & H_1 .

$$H_0: \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 - \mu_2 \neq 0$$

(2-tailed test)

② Select the significance level

$$\alpha = 0.01$$

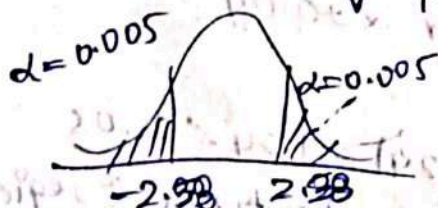
③ Find z-st values

$$Z_{st} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$= \frac{(78.5 - 77) - (0)}{\sqrt{\frac{(14)^2}{140} + \frac{(16)^2}{160}}}$$

$$= 0.866$$



$$C.V = 2.58$$

$$Z_{st} < C.V \Rightarrow 0.88 < 2.58$$

$\therefore$ lies in do not reject region.

$$④ P(0.866) = 0.805$$

$$\Rightarrow \cancel{1 - 0.805} = \cancel{0.195}$$

$\therefore \alpha = 0.05$ $[P > \alpha] \rightarrow$ we do not reject H_0
There is no diff in pres of married or unmarried woman

⑤ Who has a longer life span. Test at $\alpha = 0.05$.

Non-drug user
 $\bar{X} = 82.5y$
 $s = 12y$
 $n = 120$

drug-user
 $\bar{X} = 72.5y$
 $s = 12.5y$
 $n = 50$

Soln Define H_0 & H_1

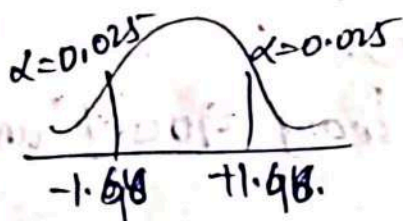
$H_0: \mu_1 - \mu_2 = 0$ | $H_1: \mu_1 - \mu_2 \neq 0$ (2 tailed)

⑤ select significant level (α)

$$\alpha = 0.05$$

③ find z-st value

$$z_{st} = \frac{(82.5 - 72.5) - 0}{\sqrt{\frac{(12)^2}{120} + \frac{(12.5)^2}{50}}} = 4.808$$



$$|c.v| = 1.68$$

$$z_{st} = 4.808$$

$z_{st} > c.v \Rightarrow$ lies in the reject region.

④ $P(4.808) = 0$, $\alpha = 0.05$

$P < \alpha \Rightarrow$ we reject null hypothesis (H_0)

i.e. There is a longer life span for Non-drug user than drug user.

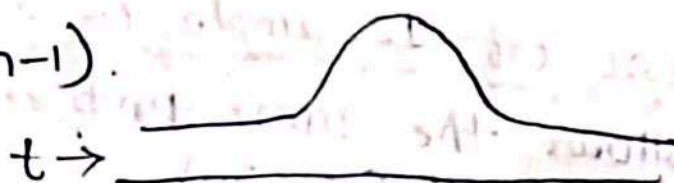
* Student's t-distribution (t-test).

* we use this when sample size (n) < 30

* Here there is unknown std devⁿ of populⁿ (σ)

* shape of the distⁿ is based on degree of

freedom ($n-1$).



tail part is heavier & we take small sample.

* For small sample $\Rightarrow$ heavier tails
for large sample $\Rightarrow$ less tails.

* Formula
$$t_{st} = \frac{\bar{X} - \mu}{\left(\frac{s}{\sqrt{n}}\right)}$$
 //

* max data covered at tail part.

Z-test	T-Test
$\rightarrow n \geq 30$	$\rightarrow n < 30$
$\rightarrow$ Known (σ)	$\rightarrow$ Unknown (σ)
$\rightarrow$ shape of dist ⁿ is based on sample	$\rightarrow$ shape of dist ⁿ is based on degree of freedom (n)
$\rightarrow z_{st} = \frac{\bar{X} - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)}$	$\rightarrow t_{st} = \frac{\bar{X} - \mu}{\left(\frac{s}{\sqrt{n}}\right)}$

* It is a member of continuous prob distⁿ.

* It is used when the sample size is small ($n < 30$) & popⁿ std devⁿ is unknown (σ).

* It was developed by "William Sealy Gosset" and the "pseudonym" student.

* t-distⁿ is equal to z-distⁿ, the shape of the distⁿ depends on degree of freedom ($n-1$)
 sample size

* If there are n -obsⁿ in the sample & ' k ' parameters are estimated from the sample then degree of freedom is " $n-k$ " ($k = k_{hi}$).

* The t-distⁿ is symmetric & bell shaped like a normal distⁿ, but it has heavier tails, i.e. it is more prone to producing values that fall far from its mean.

* Assumptions of 1-sample t-Test

1. populⁿ follows the normal prob distⁿ & sample size is small ($n < 30$).

2. popⁿ std deviation is unknown. (σ)

① A teacher claims that the avg score of her students is 70. A sample of 10 students scores are 65, 68, 75, 70, 72, 69, 74, 71, 73, 67. Test at $\alpha = 0.05$.

Sol: Find H_0 & H_1 .

$$H_0: \mu = 70$$

$$H_1: \mu \neq 70. \text{ (2-tail)}$$

② Select significant value (α)

$$\alpha = 0.05$$

③ Choose appropriate test statistic ($n < 30$, σ is unknown)

$$t_{st} = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}}\right)}$$

$$\bar{x} = \frac{65 + 68 + 75 + 70 + 72 + 69 + 74 + 71 + 73 + 67}{10}$$

$$\boxed{\bar{x} = 70.4}$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \boxed{3.21}$$

(degree of freedom = $n-1=9$)

$$t_{st} = \frac{70.4 - 70}{\left(\frac{3.21}{\sqrt{10}}\right)} = \boxed{0.39}$$

C.V for 0.39

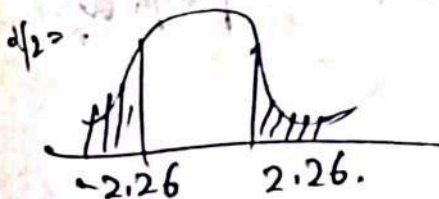
$$\boxed{= 2.262}$$

0.05

$$\boxed{0.39 < 2.26}$$

$$t_{st} < \text{C.V.}$$

df	
9	2.262



lies in do not reject region

$\therefore$ Do not reject null hypothesis (H_0).

i.e. The avg score of her students is 70.

$$C.I = \bar{x} \pm t_{(\alpha/2, df)} \left(\frac{s}{\sqrt{n}}\right) = 70.4 \pm 2.262 \left(\frac{3.21}{\sqrt{10}}\right)$$

$$C.I = [68.10, 72.7] \quad (\mu = 70)$$

$\therefore \mu$ lies in boundaries $\Rightarrow$ do not reject H_0

② A manufacturer wants to test a hypothesis to see if the diameter of their 30m bolts are being manufactured properly. Test at $\alpha = 0.05$. He collected sample of 6 and their values are 30.02, 29.99, 30.11, 29.97, 30.01, 29.99.

Sol: Define H_0 & H_1 .

$$H_0: \mu_1 = 30m$$

$$H_1: \mu_1 \neq 30m. \text{ (2-tail)}$$

② select significant value ($\alpha = 0.05$)

③ choose appropriate statistic ($n \geq 6 < 30 \rightarrow t\text{-test}$)

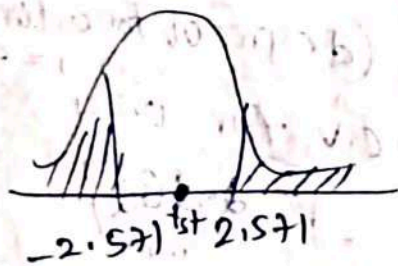
$$\bar{x} = \frac{30.02 + 29.99 + 30.11 + 29.97 + 30.01 + 29.99}{6}$$

$$= 30.015$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \frac{s^2 = 0.00274}{s = 0.0523}$$

$$t_{st} = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}}\right)} = \frac{30.015 - 30}{\left(\frac{0.0523}{\sqrt{6}}\right)} = 0.7025$$

df = 5



$$t_{st} < cv.$$

df	0.05
5	2.571

i.e. t_{st} lies in do not rejection region (H_0)

$\therefore$ The diameter of bolts = 30m are manufactured properly.

$$C.I = \bar{x} \pm t_{(\alpha/2, df)} \left[\frac{s}{\sqrt{n}} \right]$$

$$= 30.015 \pm 2.571 \left(\frac{0.0523}{\sqrt{6}} \right)$$

$$C.I = [29.96, 30.07]$$

$$(\mu = 30)$$

$\therefore \mu$ lies in the boundaries

$\therefore$ we do not reject H_0

* 2-sample t-test

- * It is used when the samples are independent.
- * If we want to estimate diff in 2 populⁿ means when std deviation of the populⁿs are unknown.

2 types of Independent 2-sample t-test:->

① 2-sample t-test with equal variance:->
(pooled variance) $(|\sigma_1^2 - \sigma_2^2| < 4) \Rightarrow$ equal variance.
 $(|\sigma_1 - \sigma_2| < 2)$

* This test is used when it can be assumed that the 2 distⁿs have the same variance ~~($\sigma_1^2 = \sigma_2^2$)~~

② 2-sample t-test with unequal variance:->
(welch's t-test) $(|\sigma_1^2 - \sigma_2^2| \geq 4) \Rightarrow$ unequal variance

* This test is also cld "welch's t-test"

* It is used only when the 2 populⁿ variances are not assumed to be equal ~~($\sigma_1^2 = \sigma_2^2$)~~.

Assumptions of 2-sample t-test with pooled variance.

1. populⁿ variances (σ_1^2 & σ_2^2) are unknown.
2. populⁿ follows the normal distribution. & sample size is $\boxed{n_1 + n_2 > 40}$.
3. The 2 samples are independent.
4. The 2 populⁿ variances are equal ($\sigma_1^2 = \sigma_2^2$)

Degree of freedom for 2 sample t-test is:->

$$df = n_1 + n_2 - 2 //$$

$$\text{std dev}^n = \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} //$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} //$$

pooled variance

$$t_{st} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} //$$

(pooled variance)

* If st devⁿ (σ) is given, convert to variance (σ^2)

Z-test

① C.I = $\bar{x} \pm Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$ — 1-test

② $Z_{st} = \frac{\bar{x} - \mu}{\left(\frac{\sigma}{\sqrt{n}} \right)}$ → 1-test

③ $Z_{st} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$ → 2-test

④ C.I = $(\bar{x}_1 - \bar{x}_2) \pm Z_{\alpha/2} \left(\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \right)$ → 2-test

① $t_{st} = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}} \right)}$ — 1-test

② C.I = $\bar{x} \pm t_{[\alpha/2, df]} \left(\frac{s}{\sqrt{n}} \right)$ — 1-test

③ $t_{st} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$ → 2-test

C.I = $(\bar{x}_1 - \bar{x}_2) \pm t_{[\alpha/2, df]} \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$

① Suppose we have 2 groups of students those who go for tutorial help & those who do not. A sample of 50 students revealed that 22 students had gone for tutorial help & the avg grade for students who received tutorial help was 78 with std devⁿ of 5 pts & the avg grade for students who didn't received help was 72 with std devⁿ of 6 pts. what is the 95% C.I on the diff b/w these 2 groups of students?

Sol: $n_1 = 22$, $n_2 = 50 - 22 = 28$

$\bar{x}_1 = 78$, $\bar{x}_2 = 72$, $s_1 = 5$, $s_2 = 6$.

$\alpha = 95\%$

① frame Null & H_1 Hypothesis

$$H_0: \mu_1 = \mu_2$$

$$\mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 \neq \mu_2$$

$$\mu_1 - \mu_2 \neq 0$$

② Select significant level $\rightarrow \alpha = 0.05$

③ choose appropriate statistic test. ($n_1, n_2 < 30 \Rightarrow t$ -test)

$$t_{st} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(22 - 1)5^2 + (28 - 1)6^2}{22 + 28 - 2}$$

$$= \frac{21 \times 25 + 27 \times 36}{48}$$

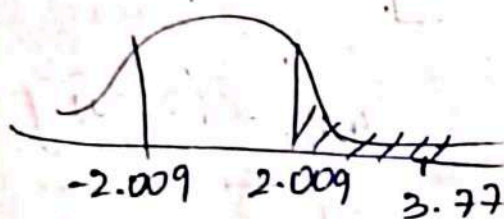
$$= 31.1875$$

$$t_{st} = \frac{(78 - 72) - (0)}{\sqrt{(31.1875) \left(\frac{1}{22} + \frac{1}{28} \right)}}$$

$$= 3.77$$

$$df = 22 + 28 - 2 = 48$$

④ C.V. = 2.009 & $t_{st} = 3.77$



$t_{st} > C.V.$

So,

t_{st} lies in rejection region.

$\therefore$ We reject the Null Hypothesis (H_0)

Let There be a diff b/w 2 groups of students //

$$C.I = (\bar{x}_1 - \bar{x}_2) \pm t_{(\alpha/2, df)} \cdot \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$= (78 - 72) \pm (2.009) \sqrt{(31.1875) \left(\frac{1}{22} + \frac{1}{28} \right)}$$

$$C.I = [2.803, 9.196]$$

② A pharmaceutical company test a new drug for improving recovery score.

→ Treatment group: mean = 8.2, std dev = 2.4, $n_1 = 20$

→ control group: mean = 6.9, std dev = 2.1, $n_2 = 22$

Test at $\alpha = 0.05$ significance level. Test whether there is a significant diff in mean recovery score b/w Treatment & control groups.

Sol: find H_0 & H_1

$$H_0: \mu_1 - \mu_2 = 0 \quad | \quad H_1: \mu_1 - \mu_2 \neq 0$$

ii) $\alpha = 0.05$

iii) $t_{st} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$ ($n_1, n_2 < 30 \Rightarrow t$ -test)

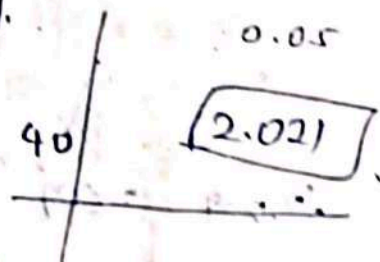
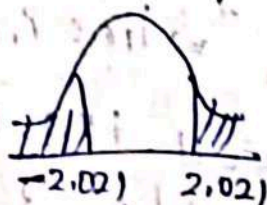
$|s_1 - s_2| = |2.4 - 2.1| = 0.3$
 $\Rightarrow$ (pooled variance) $\checkmark$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(19)(2.4)^2 + (21)(2.1)^2}{20 + 22 - 2}$$

$s_p^2 = 5.05 \Rightarrow$ Pooled variance

$$t_{st} = \frac{(8.2 - 6.9) - (0)}{\sqrt{5.05 \left(\frac{1}{20} + \frac{1}{22} \right)}} = 1.872 \quad \boxed{df = 40}$$

iv) $t_{st} < CV$



As $t_{st} < CV$, t_{st} lies in do not reject region (H_0) .

$\therefore$ We do not reject H_0 . (failed to reject H_0)

i.e. There is no significant diff in mean recovery scores b/w Treatment & control groups.

③ Two chips are tested for performance score.

chip X: $\bar{x}_1 = 58, s_1 = 6, n_1 = 12$

chip Y: $\bar{x}_2 = 50, s_2 = 6.5, n_2 = 12$

Test at $\alpha = 0.05$ sig^e level. Test whether the chips differ in mean performance

Soln : $H_0 = \mu_1 - \mu_2 = 0 \quad | \quad H_1: \mu_1 - \mu_2 \neq 0$ (If sample size ≥ 40 check $|s_1 - s_2| < 2$
 $|s_1^2 - s_2^2| < 4$)

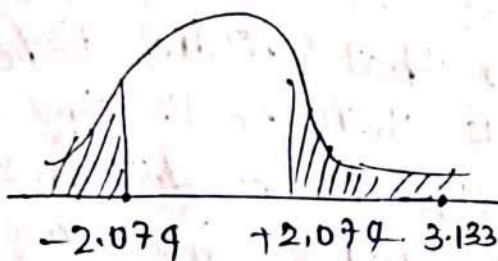
ii) $\alpha = 0.05$

iii) $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(11)(6)^2 + (11)(6.5)^2}{22}$

$= \boxed{39.125} \Rightarrow$ pooled variance.

$t_{st} = \frac{(58 - 50) - (0)}{\sqrt{39.125 \left(\frac{1}{12} + \frac{1}{12} \right)}}$

$= \frac{8}{\sqrt{39.125 \left(\frac{1}{6} \right)}} = \boxed{3.133}$



$t_{st} > cv$

df	0.05
22	cv $\boxed{2.074}$

$t_{st} > cv$, as t_{st} doesn't in cv region, so we reject H_0 .

We reject null hypothesis (H_0)

i.e. The chips differ in mean performance

As $\bar{x}_1 > \bar{x}_2$, chip X is fast & chip Y is slow.

④ A company introduce a new delivery process.

New process: $\bar{x}_1 = 120, s_1 = 10, n_1 = 35$

Old process: $\bar{x}_2 = 125, s_2 = 10, n_2 = 35$

Test at $\alpha = 0.05$. Test whether the new process is faster!

Soln $n_1 + n_2 > 40$ (or)

$|s_1 - s_2| < 2$
 $s_1 = s_2 \Rightarrow t$ -test

$H_0: \mu_0 \geq \mu_N$
 $\mu_0 \leq \mu_N$

$H_1: \mu_0 < \mu_N$ (left tail test)
 $H_1: \mu_N > \mu_0$ (right tail test)

$$\alpha = 0.05$$

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} = \frac{(34)(10)^2 + (34)(10)^2}{68} = 100 \Rightarrow \text{pooled var}$$

$$t_{st} = \frac{(125-120) - (0)}{\sqrt{100 \left(\frac{1}{35} + \frac{1}{35} \right)}} = 42.09$$

$$d.f. = 68$$

$$(2 + 66)$$

$$0.05$$

$$2.69$$

$$-1.65$$

$$1.65$$

$$2.69$$

$$-1.65$$

$$1.65$$

$$2.69$$

$$-1.65$$

$$1.65$$

$$2.69$$

$$-1.65$$

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$$2.69$$

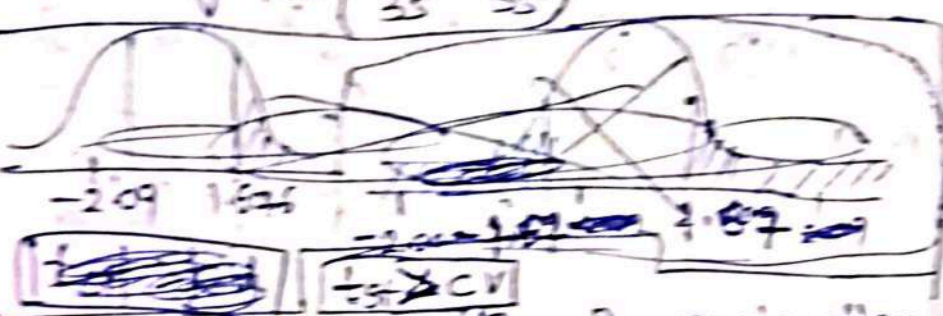
$$-1.65$$

$$1.65$$

$$2.69$$

$$-1.65$$

$$1.65$$



$\therefore t_{st} > CV$, t_{st} lies in rejection region.
 $\therefore$ We reject the null hypothesis (H_0)
 i.e. The new process is faster //

5) A company makes a claim that children (in the age group 6 to 12) who drink their health drink will grow taller than the children who don't drink that health drink. In the given table shows avg increasing ht over 1 year period from 2 groups:
 1. Drinking the health drink 2. Not drinking the health drink

Test at $\alpha = 0.05$ & test whether the increase in ht for the children who drink the health drink is atleast 1.2 cm

Exp.	s.size	Avg ht (in cm)	SD
drink health drink	80	7.6 cm	1.1 cm
donot drink health drink	80	6.3 cm	1.3 cm

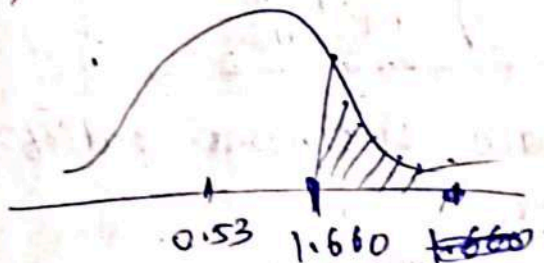
$$Sol: \rightarrow H_0: \mu_1 \geq 1.2 \text{ cm} \quad H_1: \mu_1 < 1.2 \text{ cm} \quad (\text{left tail})$$

ii) $\alpha = 0.05$

iii) $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2} = \frac{(79)(1.1)^2 + (79)(1.3)^2}{80 + 80 - 2}$
 $= 1.45$

iv) $t_{st} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{(7.6 - 6.3) - (1.2)}{\sqrt{1.45 \left(\frac{1}{80} + \frac{1}{80} \right)}}$
 $= \boxed{0.53}$

v) $df = 80 + 80 - 2 = 158$



	0.05
df	
100	1.660

$\therefore \boxed{t_{st} < CV}$

$\therefore$ we do not reject null hypothesis. (H_0)
 $\therefore$ The children who drink the health drink
is atleast 1.2cm //

(6) $\rightarrow$

* Two sample t-test with unequal variance

* Assumptions: $\Rightarrow$ (Welch's test)

1. The populⁿ variances σ_1^2, σ_2^2 are unknown.
 2. populⁿ follows the Normal distⁿ (or) $\boxed{n_1 + n_2 < 40}$
 3. The 2 samples are independent.
 4. The 2 samples variances are not equal ($\sigma_1^2 \neq \sigma_2^2$).
- The sampling distⁿ of the difference in estimated

means $\bar{x}_1 - \bar{x}_2$ follows t -dist with mean $\mu_1 - \mu_2$ & std devⁿ $S_u = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

5. Degree of freedom = $\frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}}$

6. t -statistic formula $t_{st} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$

④ A company wants to compare customer satisfaction scores b/w 2 diff stores.

Store A
mean = 80
SD = 6
 $n = 40$

Store B
75
9
35
Test at $\alpha = 0.05$
significance level
& test if there is
a significant diff
in customer satisfaction
scores b/w 2 stores

Sol: Mainly based on std devⁿ we go with test.

$|s_1 - s_2| \geq 2 \text{ (or)} |\sigma_1^2 - \sigma_2^2| > 4. \quad (6 - 9 = -3) = 3$

i) $H_0 = \mu_1 - \mu_2 = 0 \quad | \quad H_1: \mu_1 - \mu_2 \neq 0.$

ii) significant level $\alpha = 0.05$

iii) $t_{st} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{(80 - 75) - 0}{\sqrt{\frac{6^2}{40} + \frac{9^2}{35}}}$

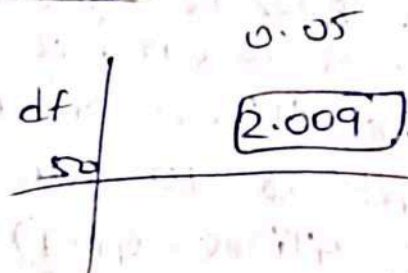
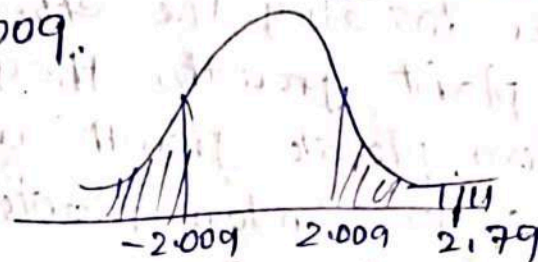
$t_{st} = 0.79$

iv) def $S_u = 1.79 = \sqrt{\left(\frac{6^2}{40} + \frac{9^2}{35}\right)} = 1.79$

$$iv) df = \left(\frac{\left(\frac{6^2}{40} + \frac{9^2}{35} \right)^2}{\frac{\left(\frac{6^2}{40} \right)^2}{39} + \frac{\left(\frac{9^2}{35} \right)^2}{34}} \right) = \left(\frac{10.33}{\frac{0.81}{39} + \frac{5.36}{34}} \right)$$

$$df = 57.89$$

$$\therefore C.V = 2.009$$



$\therefore t_{st} > CV$, t lies in rejection region.

$\therefore$ we reject Null hypothesis (H_0)

There is a significant diff in customer satisfaction scores b/w 2 stores.

Q) we want to test if there is a significant diff b/w the avg marks of boys & girls in a class.

Boys:	75	78	90	74	76	72
Girls:	68	70	65	66	69	67

(pooled variance $(s_1 - s_2) < 2$)

test at $\alpha = 0.05$.

$$\bar{x}_1 = 74.2, \bar{x}_2 = 67.5, s_1 = 2.86, s_2 = 1.87$$

Sol: $\bar{x}_1 = 74.2, \bar{x}_2 = 67.5$

$$df = 10$$

$$s_1 = 2.86, s_2 = 1.87$$

$$(s_1 - s_2) < 2 \rightarrow \text{pooled variance} \quad (2.86 - 1.87 = 0.99 < 2)$$

i) $H_0: \mu_1 = \mu_2$ | $H_1: \mu_1 \neq \mu_2$
 $\mu_1 - \mu_2 = 0$ | $\mu_1 - \mu_2 \neq 0$

ii) sig. level $\alpha = 0.05$

iii) pooled var $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$

$$= \frac{(6-1)(2.86)^2 + (6-1)(1.87)^2}{6+6-2} = 5.83$$

$$9v) t_{st} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{(14.2 - 6.15) - 0}{\sqrt{5.83 \left(\frac{1}{6} + \frac{1}{6} \right)}} = \boxed{4.806}$$

$$t_{cv} = 2.262$$

$t_{st} > t_{cv} \therefore$ we reject null hypothesis (H_0)
 $\therefore$ There is a significant diff in avg marks of boys & girls.

③ A researcher is investigating the effect of a new fertilizer on a plant growth, they measure the hts of 10 random plants growth with the new fertilizer (grp 1) & 12 random plants growth with std fertilizer (grp 2). The data is:

Grp 1	21, 16, 22, 23, 19, 24, 20, 25, 17, 17
Grp 2	23, 23, 22, 21, 26, 23, 22, 22, 24, 29, 21, 24

Can the researcher use Welch's & sample t -test to determine if there is a significant diff in avg heights of plants b/w the 2 grps. Test at $\alpha = 0.10$ significance level.

Sol: i) ~~$\bar{x}_1 = 22.58$~~ , $\bar{x}_1 = 20.4$, $\bar{x}_2 = 22.58$

$s_1 = 3.13$, $s_2 = 1.62$ ($|s_1 - s_2| = 3.13 - 1.62 = 1.51$)

i) $H_0: \mu_1 - \mu_2 = 0$ | $H_1: \mu_1 - \mu_2 \neq 0$

ii) $\alpha = 0.10$

iii) Pooled variance, $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$
 $= \frac{(10 - 1)(3.13)^2 + (12 - 1)(1.62)^2}{10 + 12 - 2}$
 $= \boxed{5.85}$

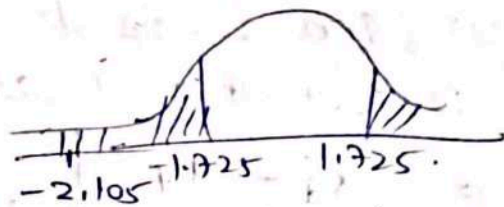
iv) $t_{st} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{(20.4 - 22.58) - 0}{\sqrt{5.85 \left(\frac{1}{10} + \frac{1}{12} \right)}} = \boxed{-2.105}$

$$df = 20.$$

$$\alpha = 0.10.$$

$$CV = 1.725$$

$$|t_{st}| = |2.105|.$$



$\therefore |t_{st}| > CV \Rightarrow$ lies in reject region

$\therefore$ we reject the null hypothesis (H_0)

$\therefore$ There is a sign. diff in the avg hts of plants b/w the 2 grps.//

* paired t-test (dependent t-test)

* The variables are dependent.
(samples)

* It is like 1 sample t-test & we test before & after of exam.

* paired t-test is a statistical test used to compare the mean's of 2 related grps (or) to see if there is a significant diff b/w them.

* paired t-test is applied when $\Rightarrow$

1. The same subjects are measured before & after a coaching./treatment.

2. Matched subjects are used.
(paired)

3. The data is continuous

4. The differences are approximately normally distributed.

* This test is used when the samples are dependent i.e. when there is only one sample that has been tested twice./ where there are 2 samples that have been paired (matched).

examples:- 1. sugar levels before & after eating.

2. Heart rate before & drinking a cup of coffee after

3. Alcohol consumption before & after marriage or breakup.

4. Maths scores before & after attending the course.

5. Test scores based on age groups.

* It is just like 1 sample t-test.

$$t_{st} = \frac{\bar{X}_d - \mu_d}{S_d / \sqrt{n}}$$

n = paired values.

$\bar{X}_d = \bar{X}_1 - \bar{X}_2$
 diff b/w 2 sample mean
 $S_d =$ diff b/w 2 sample std dev

① A company gives a six employees a training program & records of their test scores before & after training. Test at $\alpha = 0.05$.

Before	50	60	55	58	62	53
After	55	63	54	60	65	57
diff	-5	-3	1	-2	-3	-4

Sol: i) $H_0: \mu_d = 0$

$H_1: \mu_d \neq 0$

ii) $\alpha = 0.05$

iii) $\bar{X}_1 = 56.33, \bar{X}_2 = 59$

$\therefore \bar{X}_d = -2.67$

$t_{st} = \frac{-2.67 - 0}{(S_d / \sqrt{n})}$

~~$S_1 = 9.15$~~
 ~~$S_2 = 9.43$~~

$S_d = 2.107$

$S_1 = \sqrt{\frac{\sum (X_1 - \bar{X}_1)^2}{n-1}}$

$S_d = \sqrt{\frac{\sum (d_i - \bar{d})^2}{n-1}}$

$t_{st} = \frac{-2.67}{\left(\frac{2.107}{\sqrt{6}}\right)} = -3.16$

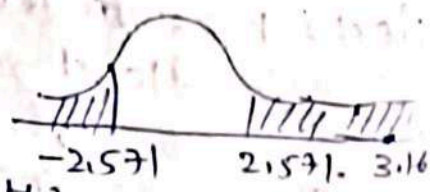
$t_{(\alpha, df)} = CV = t_{(0.05, 5)} = 2.571$

$df = n-1 = 6-1 = 5$

$|t_{st}| > CV \Rightarrow$ Res in rejn region

(check for 2-tail test)

$\therefore$ we reject null hypothesis (H_0)



i.e. There is a diff b/w the scores of employees before & after training.

② Suppose 5 peoples weights are measured before & after a 1 month diet

Before	80	75	90	85	95
After	78	74	85	83	92

Test at $\alpha = 0.05$

So $H_0: \mu_d = 0$ | $H_1: \mu_d \neq 0$ ii) $\alpha = 0.05$

df	2	1	5	2	3
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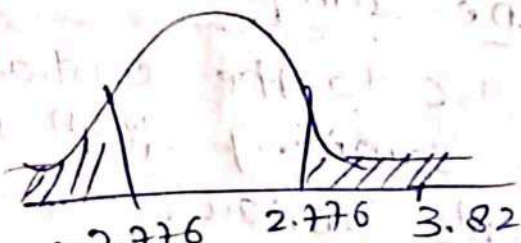
$$\bar{x}_d = 2.6$$

$$s_d = \sqrt{2.3} = 1.52$$

$$\text{iii) } t_{st} = \frac{2.6 - 0}{\left(\frac{1.52}{\sqrt{5}}\right)} = 3.82$$

$$df = 5 - 1 = 4$$

$$t_{[0.05, 4]} = 2.776$$



$\therefore t_{st} > cv$ $\therefore$ lies in rejection region

$\therefore$ We reject null hypothesis (H_0)

$\therefore$ There is a diff in weights before & after a 1 month diet //

③ A researcher believes that people drink more coffee on Monday's, than other days of the week. Based on the sample of 50 coffee drinkers the mean diff was estimated as 14 ml & the corresponding std devⁿ was 8.5 ml. conduct an appropriate hypothesis test at $\alpha = 0.1$ to check the claim that people drink on avg 10 ml more coffee on Monday's compared to other days of the week.

So $H_0: \mu_d \leq 10 \text{ ml}$ | $H_1: \mu_d > 10 \text{ ml}$

(right-tailed)

$$\bar{x}_d = 14 \text{ ml}, s_d = 8.5 \text{ ml}$$

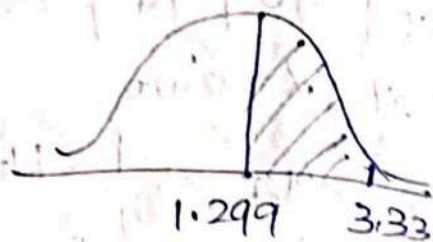
ii) $\alpha = 0.1$

H_0	H_1
$\leq$	$>$
$\geq$	$<$

$$iii) t_{st} = \frac{\bar{x}_d - \mu_d}{(s_d/\sqrt{n})} = \frac{14 - 10}{\left(\frac{8.5}{\sqrt{50}}\right)} = \boxed{3.33}$$

$$iv) t_{[0.1, 49]} = \boxed{1.299}$$

$\therefore t_{st} > t_{cv} \therefore$ lies in rejection region



$\therefore$ we reject null hypothesis (H_0)

$\therefore$ people drink on avg 10 ml more coffee on mondays compared to other days of the week.

* Hypothesis Test for population proportion (p)

* one-sample Test for proportion ($\hat{p}$)

* Acc to the central limit Theorem of proportion the sampling distⁿ of proportions ($\hat{p}$) for a large sample follows an approximate Normal distⁿ with mean (p) [popⁿ proprⁿ] & std deviation is p.e

$$\sqrt{\frac{p(1-p)}{n}}$$

i.e. the z-static as defined below will follow a std normal dist

$$\text{std dev} = \sqrt{\frac{p(1-p)}{n}}$$

$$Z_{st} = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

[$\hat{p} \rightarrow$ instead of $\bar{x}$
 $p \rightarrow \mu$]

* If $n > 30 \rightarrow$ we do Z_{st}
 $n < 30 \rightarrow$ we do t_{st}

Ex: ① $\rightarrow$ Acc to a study exactly 18% of gift cards purchased from E-commerce portals are never used. The manager of an E-commerce company wanted to test whether the claim is true. she collected the data of 250 gift card purchased & found that 22 gift cards

do not used till its expiry date.

① conduct an appropriate hypothesis test at 5% significance to check whether the claim that exactly 12% gift cards are never used is true or not.

② calculate the 95% C.I for the proportion of gift cards that are not used.

Soln: ① frame H_0 & H_1 .

$$H_0: p = 12\% \\ p = 0.12$$

$$H_1: p \neq 12\%$$

$$H_1: p \neq 12\% \\ p \neq 0.12 \text{ (2 tail)}$$

② $\alpha = 0.05$.

③ chose appropriate statistic (Z_{st} b/c $n > 30$).

$$Z_{st} = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

$$\hat{p} = \frac{22}{250} = 0.088$$

$$Z_{st} = \frac{0.088 - 0.12}{\sqrt{\frac{0.12(0.88)}{250}}} = -1.56$$

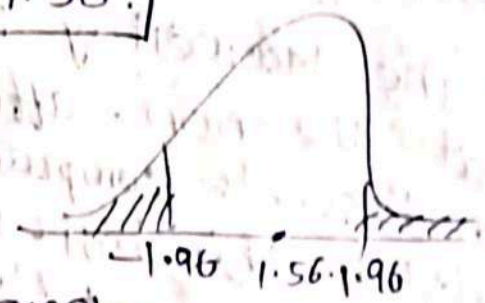
$$\therefore |Z_{st}| = 1.56 < |CV|$$

$$\therefore |Z_{st}| < |CV|$$

$\therefore Z_{st}$ lies in do not reject region

i.e. Z_{st} value is not the part of the reject region, so we fail to reject H_0 . / do not reject H_0 .

i.e. exactly 12% gift cards are never used is true.



$$④ \text{ C.I} = \hat{p} \pm Z_{(\alpha/2)} \left(\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right) \quad (n=250)$$

$$\text{C.I} = 0.088 \pm 1.96 \left(\sqrt{\frac{0.088(0.912)}{250}} \right)$$

$$\text{C.I} = [-0.03, 0.206]$$

$\therefore p = 0.12$ is in b/w for

* The proporⁿ of unused is likely to lie b/w $[-0.03, 0.2]$, so we do not reject H_0 //

∴ 12% gifts are never used //

* 2-sample Test for proportion

* when the proportions are estimated from a large sample then sampling distⁿ of proportions follow a normal distⁿ acc to the central limit Theorem.

* let $\hat{p}_1$ & $\hat{p}_2$ be the estimated values of proportions from large samples.

* The diff $(\hat{p}_1 - \hat{p}_2)$ is the hypothesized difference in populⁿ proportions.

$$Z_{st} = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$\hat{p} = \frac{n_1 \cdot \hat{p}_1 + n_2 \cdot \hat{p}_2}{n_1 + n_2}$$

Here $\hat{p}$ is the pooled estimate for proportion

① The Marketing Mgr of a company believes that the non-affluent customers are sensitive to discounts compared to affluent customers. To validate this hypothesis discount coupons were sent to non-affluent & affluent customers. & the data is given below: use an appropriate hypothesis to check whether there is any diff in proportion of customers who use discount coupons at $\alpha = 0.05$.

Group	sample size	no. of customers Using discount coupons	Estimated prop ⁿ
Non-Affluent	500	145	0.29
Affluent	300	42	0.14

Sol: i) $H_0 = p_1 = p_2$ | $H_1: p_1 \neq p_2$

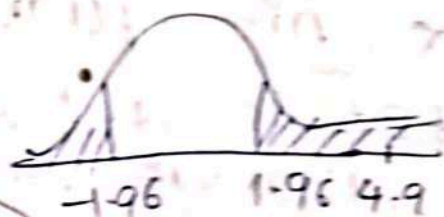
ii) $\alpha = 0.05$

$$iii) \hat{p}_1 = \frac{145}{500} = 0.29, \hat{p}_2 = 0.14$$

$$\hat{p} = \frac{500(0.29) + (300)(0.14)}{500+300} = \boxed{0.233}$$

$$Z_{st} = \frac{(0.29 - 0.14) - 0}{\sqrt{0.233(1-0.233)\left(\frac{1}{500} + \frac{1}{300}\right)}} = \boxed{4.9}$$

$$iv) = \boxed{Z_{st} > CV} \Rightarrow \boxed{P=0. (P < \alpha)} \Rightarrow \boxed{0 < 0.05}$$



$\therefore Z_{st}$ lies in rejection region
 $\therefore$ we reject Null hypothesis (H_0).

$\therefore$ There is a diff in proportion of customers who use discount coupons //.

* Parametric & Non-parametric Test

Parametric Test	Non-parametric test
1. It makes assumptions about parametric distn. (Normal distn).	1. It makes no assumptions abt non-parametric test. Here distn is free (free distn test).
2. It requires Interval or ratio data. i.e data is continuous & measurable Ex: Ht, wt, Marks.	2. It works with ordinal or Nominal data. i.e Data is categorical (or) Ranks / satisfaction levels. Ex: Examples are $\rightarrow$ chi-square test.
3. Examples are z-test, T-test & ANOVA & Pearson correlation.	4. Efficiency is less powerful but more flexible when assumptions are violated.
4. Efficiency is more powerful when assumptions are met.	5. Here sample size is smaller.
5. Here sample size is larger.	

6. Distⁿ is symmetric

7. We need summary statistics about population. ~~pop~~ mean, std devⁿ.

6. Distⁿ is not symmetric (skewed)

7. We don't need summary statistics. It is not about the popⁿ parameters.

* Chi-Square Test

* It is used to determine the relⁿship b/w categorical variables.

* It is a non-symmetric & non-parametric test.

* chi-square distⁿ will have only the values (from 0 to ∞)

* chi-square distⁿ depends on degree of freedom

* If the df changes then distⁿ also change.

* $df = n - 1$: n = no. of categories

* There are 2 types of chi-square test

1. chi-square Test for goodness of fit Test.
2. chi-square Test for Independence.

* It is indicated with " χ^2 ". (given)

$$\chi^2_s = \sum_{i=1}^n \left(\frac{O_i - E_i}{E_i} \right)^2$$

Statistic formula.

$O \rightarrow$ observed freq values in category
 $E \rightarrow$ Expected freq values in category.
(we need to find out)

* Conclusion is:

i) $\chi^2_s > CV \Rightarrow$ reject H_0

ii) $\chi^2_s < CV \Rightarrow$ Do not reject H_0 .

① Hanuman Airline (HA), operated daily flights to several Indian cities. One of the problems HA faces is the food preferences by the passengers. "Captain Cook" that oper^s Mgr of HA believes that 35% of their passengers

prefer vegetarian food & 40% prefer non-veg food & 20% low calorie food & 5% request for diabetic food. A sample of 500 passengers were chosen to analyze the food preferences and the data is given below, conduct a chi-square test to check whether captain cook's belief is true at $\alpha = 0.05$.

Food Type. $\Rightarrow$	veg	Non veg	low calories	diabetic.
no. of passengers $\Rightarrow$	190	185	90	35

Solⁿ i) H_0 $\Rightarrow$
prob distⁿ of the food preferences is same.
(mentioned in qn)

H_1 $\Rightarrow$ prob distⁿ of the food preferences is not same.

ii) $\alpha = 0.05$.

iii) choose an appropriate Test (based on categories, we choose chi-sq test)

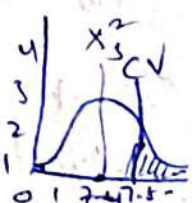
*The given values are observed values

Observed values	190	185	90	35
Expected values	175	200	100	25

(500x35%) (500x40%) (500x20%) (500x5%)

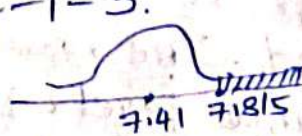
$$\chi^2 = \frac{(15)^2}{175} + \frac{(15)^2}{200} + \frac{(10)^2}{100} + \frac{(10)^2}{25} = \boxed{7.4}$$

(1.285) + (1.125) + (1) + (4)



iv) $df = n - 1 = 4 - 1 = 3$.

$CV = 7.815$



df	0.05
3	7.815

$\therefore \chi^2 < CV \Rightarrow$ we do not reject H_0 .
i.e. The prob distⁿ of the food preferences is same.

We conclude that captain cook's belief abt the food preferences of HA is true.

Q A die was rolled 120 times observed face values counts are 18, 22, 20, 16, 24, 20. Test at $\alpha = 0.05$

Sol: H_0 : Die is fair | H_1 : Die is Unfair

$$\boxed{\alpha = 0.05}$$

iii) Expected value = $120 \times \frac{1}{6} = \boxed{20}$

$$\therefore \chi^2 = \frac{(18-20)^2}{20} + \dots + \frac{(20-20)^2}{20}$$

$$\boxed{\chi^2 = 2}$$

$$df = 6 - 1 = 5$$

$$\boxed{CV = 11.070}$$

$\boxed{\chi^2 < CV} \rightarrow$ we do not reject H_0

$\therefore$ The Die is fair //

* These are goodness of fit ~~test~~ problems.

H_0 is not rejected $\rightarrow$ valid fitness.

H_0 is rejected $\rightarrow$ Invalid fitness.

Q A candy company claims that its bags of mixed fruit chews contain an equal no. of 5 flavours apple, lime, cherry, orange, grape. A random sample of 10 bags is taken and the no. of each flavour is counted. The results are shown in table at 5% significance level. Does the data provide evidence that the flavour distn is diff from the company's claim.

flavour	Apple	lime	cherry	Orange	Grape
observed count.	180	250	120	225	225

Sol: i) Frame H_0 & H_1 .

H_0 : The proportion of each flavour in the populⁿ of candy bags is equal.

H_1 : The proportion of atleast one of the flavour is diff from the candy bags from company's claim.
 $\therefore$ (or) is not equal

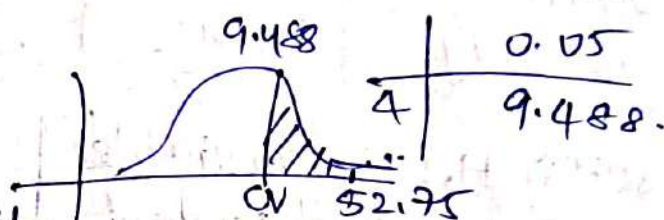
ii) $\alpha = 0.05$

iii) Expected values. = $\frac{180+250+120+225+225}{5} = \frac{1000}{5} = 200$
 $= [200]$

$$\chi^2 = \frac{(20)^2}{200} + \frac{(50)^2}{200} + \frac{(80)^2}{200} + \frac{(25)^2}{200} + \frac{(25)^2}{200}$$

$\chi^2 = 52.75$

iv) $df = n - 1 = 5 - 1 = 4$



$\because \chi^2 > CV \Rightarrow$ we reject H_0

$\therefore$ The proportⁿ of each flvr in the populⁿ of candy bags $\neq$
 * Chi-Square Test for Independence not equal

(Independence Test / Test of Association)

* It is a non-parametric test used to determine whether your data is significantly diff from what you expected.

* It determines whether a 2 categorical variables are related or not. (dependent / independent).

* If 2 variables are related, the probability of 1 variable having a certain value is dependent on the value of other variable.

* ~~Here~~ expected values ~~are~~
 To find the

$$\text{Expected freq (E)} = \frac{\text{row total} * \text{col total}}{N}$$

where N = Grand total.

* chi-square st $\Rightarrow \chi^2 = \sum_{i=1}^n \left(\frac{(O_i - E_i)^2}{E_i} \right)$

* Here $df = (r-1) * (c-1)$

* Here both variables are categorical & observed must be independent.

* Independence test is used to test association b/w 2 categorical variables.

① A movie Theatre manager wants to know if there is a relationship b/w the type of movie people watch & whether they buy a snack. The manager collects data from 200 movie goers. Test at $\alpha = 0.05$

Type of movie	snacks	no-snacks
Action	60	40
comedy	30	70

Sol: i) H_0 : Two variables are independent (there is no association)

H_1 : Two variables are dependent (there is association)

ii) $\alpha = 0.05$

iii) $E = \frac{\text{row total} \times \text{col total}}{N}$

	60	40	Total
30	70		100
Total	90	110	N=200

$$R_1C_1(60) = \frac{100 \times 90}{200} = 45$$

$$R_1C_2(40) = \frac{100 \times 110}{200} = 55$$

$$R_2C_1(30) = \frac{100 \times 90}{200} = 45$$

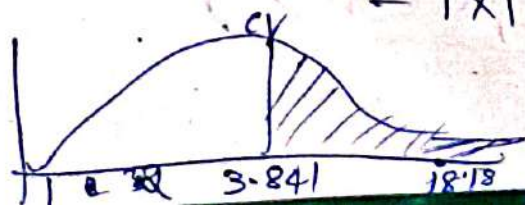
$$R_2C_2(70) = \frac{100 \times 110}{200} = 55$$

$$\chi^2 = \frac{(60-45)^2}{45} + \frac{(40-55)^2}{55} + \frac{(30-45)^2}{45} + \frac{(70-55)^2}{55}$$

$$\chi^2 = 18.18$$

$$df = (r-1)(c-1) = (2-1)(2-1) = 1 \times 1 = 1$$

if	$\alpha = 0.05$
1	3.841



$n[X^2 > CV] \rightarrow$ we reject H_0 .

$\therefore$ Two variables are dependent (there is association) //.

② The table below shows the avg no. of hrs students spend studying for class each day in a high school. Is the avg no of hrs dependent on the type of student. Test at $\alpha = 0.05$.

Hours	0-2 hrs	3-4 hrs	4-6 hrs
fresh	76	143	91
senior	147	109	64

Soln H_0 : Avg no of hrs all independent on the type of student / hrs & hrs spending time

H_1 : hrs & hrs spending time is not equal

ii) $\alpha = 0.05$ is equal //

76	143	91	310
147	109	64	320
223	252	155	N= 630

$$R_{11}(76) = \frac{223 \times 310}{630} = 109.7$$

$$R_{12}(143) = \frac{310 \times 252}{630} = 124$$

$$R_{13}(91) = \frac{155 \times 310}{630} = 76.3$$

$$R_{21}(147) = \frac{223 \times 320}{630} = 113.3$$

$$R_{22}(109) = \frac{252 \times 320}{630} = 128$$

$$R_{23}(64) = \frac{155 \times 320}{630} = 78.7$$

$$X^2 = \frac{(33.7)^2}{109.7} + \frac{(19)^2}{124}$$

$$+ \frac{(14.7)^2}{76.3} + \frac{(33.7)^2}{113.3}$$

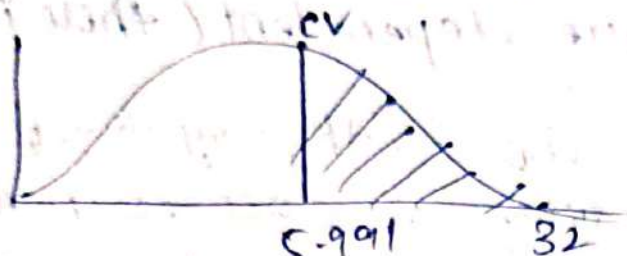
$$= \frac{(19)^2}{128} + \frac{(14.7)^2}{78.7}$$

$$= 31.7$$

$$= \boxed{32}$$

(Take round fig)

$$iv) \chi^2 = 32, \quad df = (2-1)(3-1) = 1 \times 2 = \boxed{2}$$



$$\begin{array}{c|c} & 0.05 \\ \hline 2 & \boxed{5.991} \end{array}$$

$\chi^2 > cv \Rightarrow$ we reject H_0 .

$\therefore$ snrs & Inrs spending time is not equal

③ 500 Elementary school boys & girls are asked what is their favt color: Blue, green, pink.

	Blue	Green	pink
Boys	100	150	20
Girls	20	30	180

Test at $\alpha = 0.05$

would you conclude

that there is a relⁿship b/w genders & favt color.

Sol:- H_0 : There is no relⁿship b/w gender & favt color.

H_1 : There is a relⁿship b/w gender & favt color.

$$\alpha = 0.05$$

100	150	20	270
20	30	180	230
120	180	200	500

$$R_1 C_1 (100) = \frac{120 \times 270}{500} = 65$$

$$R_2 C_3 (180) = \frac{200 \times 220}{500} = 88$$

$$R_1 C_2 (150) = \frac{180 \times 270}{500} = 97$$

$$R_1 C_3 (20) = \frac{200 \times 270}{500} = 108$$

$$R_2 C_1 (20) = \frac{120 \times 230}{500} = 55$$

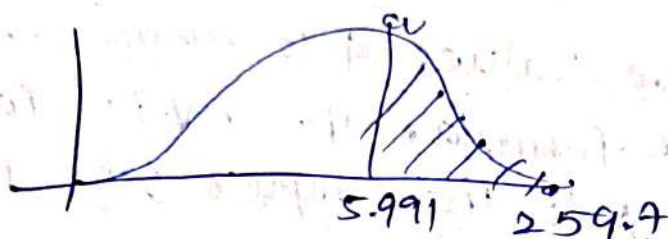
$$R_2 C_2 (30) = \frac{180 \times 230}{500} = 83$$

$$iv) \chi^2 = \frac{(35)^2}{65} + \frac{(53)^2}{97} + \frac{(88)^2}{108} + \frac{(35)^2}{55} + \frac{(53)^2}{83} + \frac{(88)^2}{92}$$

$$\boxed{\chi^2 = 259.7}$$

$$df = (2-1)(3-1) = 1 \times 2 = 2$$

	0.05
2	5.991



$$\boxed{\chi^2 > cv} \Rightarrow \text{we reject } H_0$$

∴ There is a relationship b/w gender & foot color//.

* F-Test

* Z-Test & t-test are used to compare 2 mean values. ($\mu_1 = \mu_2$)

* F-Test is used to check the 2 normal populⁿs have same variance or not ($\sigma_1^2 = \sigma_2^2$)

∴ Here, one which has less variance is good work, i.e. performs very well & one which has more variance is bad work, i.e. performs bad.

∴ F-Test is developed by "Ronald Fisher"

∴ F-distⁿ was to study the behaviour of 2 variances from random samples taken from 2 independent normal populⁿs.

∴ F-distⁿ is skewed to right side, It is not a symmetric.

* Let u_1 & u_2 are chi-square random variables with d_1 & d_2 degrees of freedom respectively. then if u_1 & u_2 are independent then:

$$X = \frac{u_1/d_1}{u_2/d_2} \quad \text{when } u_1 \text{ \& } u_2 \text{ are independent.}$$

Then the prob. density fⁿ for 'x' is given by:-

$$f(x) = \begin{cases} \frac{\gamma\left(\frac{d_1+d_2}{2}\right)\left(\frac{d_1}{d_2}\right)^{d_1/2} x^{\frac{d_1}{2}-1}}{\gamma\left(\frac{d_1}{2}\right)\gamma\left(\frac{d_2}{2}\right)\left(1+\frac{d_1 x}{d_2}\right)^{\frac{d_1+d_2}{2}}} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

* F-distⁿ has +ve values & is always +ve or $x \geq 0$.

* Using the above formula, the CV is found but we do not integrate the complex fⁿs. Instead of this, we will use f-table.

* The No. of f-tables are based on α values
 $(\alpha=0.01 - 1 \text{ table}, \alpha=0.05 - 1 \text{ table} \dots)$
 $\alpha=0.05$
 $\alpha=0.01$ - 1 table

* The random variables of the F-distⁿ may also be written as:-

$$X = \left(\frac{s_1^2}{\sigma_1^2} \right) \div \left(\frac{s_2^2}{\sigma_2^2} \right) \Rightarrow X = \frac{\left(\frac{s_1^2}{\sigma_1^2} \right)}{\left(\frac{s_2^2}{\sigma_2^2} \right)}$$

$$s_1^2 = \frac{s_1^2}{d_1}, \quad s_2^2 = \frac{s_2^2}{d_2}$$

$$s_1^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

(d₁) numerator df = n₁ - 1

(d₂) denominator df = n₂ - 1

$$F_{st} = \frac{s_1^2}{s_2^2} = \frac{\text{variance 1}}{\text{variance 2}} \quad (s_1^2 > s_2^2)$$

* F-distⁿ is used in ANOVA test to test the mean values of multiple groups.

* F-Test can be used to know 2 samples from populⁿs with equal variance (or) diff variance.

① The dataset contains 480 ceramic strength measurements for 2 batches of material. The summary statistics for each batch are shown below.

Batch-1

$$n_1 = 240$$

$$\text{mean} = 688.99$$

$$s_1 = 65.5$$

Batch-2

$$n_2 = 240$$

$$\text{mean} = 611.15$$

$$s_2 = 61.8$$

Test the variances for the 2 batches are equal.
Test at $\alpha = 0.05$.

Sol: ① $H_0: \sigma_1^2 = \sigma_2^2$ (Populⁿ) | $H_1: \sigma_1^2 \neq \sigma_2^2$ (Populⁿ)

② $\alpha = 0.05$

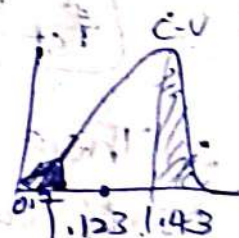
③ $f_{st} = \frac{s_1^2}{s_2^2} = \frac{(65.5)^2}{(61.8)^2} = 1.123$

find $df = ?$

$$d_1 = n_1 - 1 = 239 \text{ (numerator df)}$$

$$d_2 = n_2 - 1 = 239 \text{ (denominator df)}$$

C.V = $f(1 - \alpha/2, df_1, df_2) \rightarrow \text{Right tail}$
 $CV = \frac{1}{f(1 - \alpha/2, df_1, df_2)} \rightarrow \text{left tail}$



For 2 tails: $\alpha = \frac{\alpha}{2} = 0.025$

$CV = 1.43$

see this table.

	0.025
1	120
120	1.43

$f_{st} < CV$

left = $\frac{1}{1.43} = 0.7$

∴ We failed to reject H_0 .

∴ Variances for the 2 batches are equal.

② Is there a diff b/w the variances of the No. of weeks on the best seller list for non-fiction & fiction books. 15 New York Times best

selling fiction books had std devⁿ of 6.17 weeks on the list. 16 New York Times best selling non-fiction books had a std devⁿ of 13.12 weeks at 10% significance level, can we conclude, there is a diff in the variance.

Sol: ① $H_0: \sigma_1^2 = \sigma_2^2$ | $H_1: \sigma_1^2 \neq \sigma_2^2$ (popⁿ)

② $\alpha = 0.10$

(always $(s_1)^2 > (s_2)^2$)

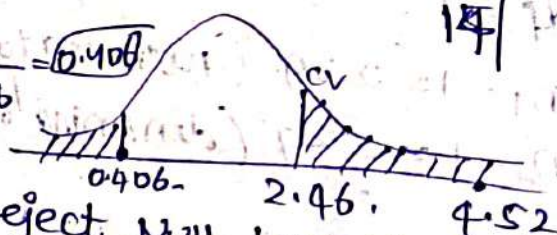
③ $f_{st} = \frac{(s_1)^2}{s_2^2} = \frac{(6.17)^2}{(13.12)^2} = \frac{38.0689}{172.1344} = 0.221$ $\frac{(13.12)^2}{(6.17)^2} = 4.52$

④. df. $\Rightarrow$ numerator df (d_1) = $n_1 - 1 = 16 - 1 = 15$
denominator df (d_2) = $n_2 - 1 = 15 - 1 = 14$

⑤ $cv = ?$ check in $\frac{\alpha}{2} = 0.05$

$CV = 2.46$ $\Rightarrow$ right side.

left = $\frac{1}{2.46} = 0.406$



	15
14	2.46

$\therefore f_{st} > CV \Rightarrow$ we reject H_0

$\therefore$ There is a diff. b/w the variances of the no. of weeks on the best seller list for non-fiction & fiction books.

* ANOVA (Analysis of variance)

* It is Analysis of variance.

* ANOVA stands for "Analysis of Variance"

* ANOVA is a statistical test used to compare the means of 3 or more independent groups to see if at least 1 group mean is significantly diff from the others.

* Why we need Anova test $\Rightarrow$ means, it is alternative to the multiple t-test.

* If we want to compare 3 grps, we could do multiple t-test (grp A v/s grp B; grp B v/s grp C; grp C v/s grp A).

But, this will increase the chance of type-1 error.

* ANOVA test is used to test all grps at a time while keeping the error rate under control.

* ANOVA is 2 types:

1. one-way ANOVA

2. two-way ANOVA

* ONE-WAY ANOVA: It is a hypothesis test in which ^{only} single factor is taken into consideration.

* TWO-WAY ANOVA: It examines the effect of 2 independent factors.

① One-way ANOVA

* It is a technique that can be used to compare means of 2/more samples.

* This technique can be used only for 1 factor (one variable). i.e. 1 way.

* If we want to study the effect of 3 levels (A_1, A_2, A_3) of a fertilizer on a plant growth where A_1, A_2, A_3 are 3 levels of the 1 factor (fertilizer - 1 factor).

* Assumptions of one-way ANOVA

① Independence of observations. i.e. the data collected from diff grps should be independent.
Ex: Scores of students in 1 class shouldn't influence scores of another class.

② Normality.

→ The data in each grp should be approximately normally distributed.

③ Homogeneity of variance. (equal variance)

→ The variance in each grp should be same.

$$[\sigma_1^2 = \sigma_2^2 = \sigma_3^2]$$

* steps to solve ANOVA Test

1. frame Null & Alternative Hypothesis
 H_0 = All grp means are equal. ($\mu_1 = \mu_2 = \mu_3$)
 H_1 = At least 1 grp mean is different / All means are different ($\mu_1 \neq \mu_2 \neq \mu_3$)
2. choose the significance level.
3. Calculate grp means & overall mean / grand mean.
4. compute the sum of squares of total / Total sum of squares
i.e. $S.S.T \Rightarrow$ total sum of squares

$$SST = \sum (x_i - \bar{x})^2 \Rightarrow SST = SSB + SSW$$

* $SSB \Rightarrow$ sum of squares for between groups.

$$SSB = \sum n_i (\bar{x}_i - \bar{x})^2$$

* $SSW \Rightarrow$ sum of squares for within groups

$$SSW = \sum (x_i - \bar{x})^2 \rightarrow \text{within groups}$$

5. compute degrees of freedom.

$$df_B (\text{df for B/w groups}) = K - 1 \quad (K = \text{no. of groups})$$

$$df_W (\text{df for within groups}) = n - K \quad (n = \text{no. of obs}^n, K = \text{no of groups})$$

6. compute the mean squares

MS_B (mean sq. for B/w groups)

$$MS_B = \frac{SSB}{df_B} = \frac{\sum n_i (\bar{x}_i - \bar{x})^2}{K - 1}$$

MS_W (mean sq for within groups)

$$MS_W = \frac{SSW}{df_W} = \frac{\sum (x_i - \bar{x})^2}{n - K}$$

7. find the F-statistic value.

$$F_{st} = \frac{MS_B}{MS_W} = \frac{\text{B/w grp variance}}{\text{within grp variance}}$$

8. Compare F_{st} value with $F_{critical}$ value.

9. Conclusion: $\left\{ \begin{array}{l} F_{st} \geq F_{cv} \Rightarrow \text{Reject } H_0 \\ F_{st} < F_{cv} \Rightarrow \text{Do not Reject } H_0 \end{array} \right\}$

① consider an exp to study the effect of 3 diff levels of a factor on a response (3 levels of a fertilizer on a plant growth). If we had 6 obs^{ns} for each level, we could write the outcome of the experiment in a table like this where A_1, A_2, A_3 are the 3 levels of the factor being studied.

A_1	A_2	A_3
6-5	8-9	13-10
8-5	12-9	19-10
4-5	9-9	11-10
5-5	11-9	8-10
3-5	6-9	7-10
4-5	8-9	12-10

Test at $\alpha = 0.05$.

Q1: 1) $H_0 = \mu_0 = \mu_1 = \mu_2$ | $H_1: \mu_0 \neq \mu_1 \neq \mu_2$

ii) $\alpha = 0.05$

iii) $\left. \begin{array}{l} A_1 \Rightarrow 5 (\mu_0) \\ A_2 \Rightarrow 9 (\mu_1) \\ A_3 \Rightarrow 10 (\mu_2) \end{array} \right\} \text{grp means}$

Overall mean = $\frac{\mu_0 + \mu_1 + \mu_2}{3} = 8. (\bar{x})$

iv) $SSB = \sum n_i (x_i - \bar{x})^2 = 6(5-8)^2 + 6(9-8)^2 + 6(10-8)^2$

$$SSB = 84$$

$$SSW = \sum (x_i - \bar{x})^2 = (5-8)^2 + (9-8)^2 + (10-8)^2$$

$$= 9 + 1 + 4 = 14$$

$$df_B = K - 1 = 3 - 1 = 2$$

$$df_W = n - K = 18 - 3 = 15$$

(6+6+6)

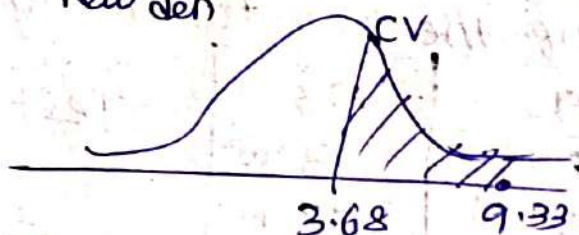
$$1 + 9 + 1 + 0 + 4 + 1 + 1 + 9 + 0 + 4 + 16 + 1 + 9 + 1 + 7 + 1 + 4 + 9 + 4 = 68$$

$$vi) MSB = \frac{SSB}{df_B} = \frac{89}{2} = 44.5$$

$$MSW = \frac{SSW}{df_W} = \frac{68}{15} = 4.53$$

$$vii) F_{st} = \frac{MSB}{MSW} = \frac{44.5}{4.53} = 9.82$$

$$F_{cv} = F(0.05, 2, 15) = 3.68$$



$\alpha = 0.05$

df_B	df_W
2	15
2	15
2	15

$\therefore F_{st} > F_{cv} \Rightarrow$ we reject null Hypothesis (H_0)

$\therefore \mu_1 \neq \mu_2 \neq \mu_3$

All means are not equal (different)

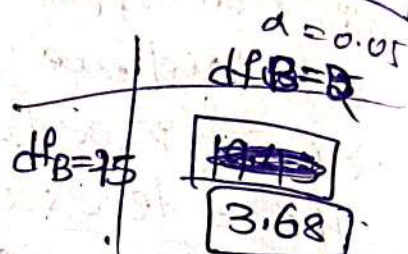
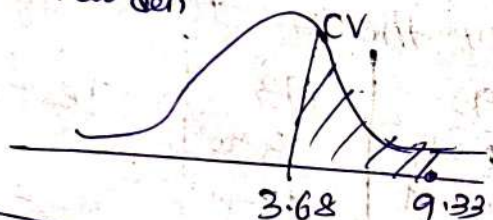
$$vi) MSB = \frac{SSB}{df_B} = \frac{84}{2} = 42$$

$$MSW = \frac{SSW}{df_W} = \frac{68}{15} = 4.5$$

$$vii) F_{st} = \frac{MSB}{MSW} = \frac{42}{4.5} = 9.33$$

$$F_{cv} = F(0.05, 2, 15) = 3.68$$

new den



$\therefore F_{st} > F_{cv} \Rightarrow$ we reject Null Hypothesis (H_0)

$$\therefore \mu_1 \neq \mu_2 \neq \mu_3$$

All means are not equal (different)

⑤ Using the following data, perform 1-way ANOVA Test at $\alpha = 0.05$.

grp 1	grp 2	grp 3
51	23	56
45	43	76
33	23	74
45	43	87
67	45	56

Sol: i) H_0 : All means are equal ($\mu_0 = \mu_1 = \mu_2$) $\mid$ H_1 : All means are not equal ($\mu_1 \neq \mu_2 \neq \mu_3$)

$$ii) \alpha = 0.05$$

iii) Grp Means:

$$G_1: 48.2$$

$$G_2: 35.4$$

$$G_3: 69.8$$

$$\text{Overall mean} = \bar{X} = \frac{\bar{X}_1 + \bar{X}_2 + \bar{X}_3}{3} = 51.13$$

$$v) SSB = 5(48.2 - 51.13)^2 + 5(35.4 - 51.13)^2 + 5(69.8 - 51.13)^2$$

$$= \boxed{3022.3495}$$

$$df_B = 3 - 1 = \boxed{2}$$

$$MSB = \frac{3022.3495}{2} = \boxed{1511.2}$$

$$vi) SSW = (2.8)^2 + (3.2)^2 + (15.2)^2 + (3.2)^2 + (18.8)^2 + (12.4)^2 + (7.6)^2 + (12.4)^2 + (7.6)^2 + (9.6)^2 + (13.8)^2 + (6.2)^2 + (4.2)^2 + (17.2)^2 + (13.8)^2$$

$$= \boxed{1860.8}$$

$$df_W = 15 - 3 = \boxed{12}$$

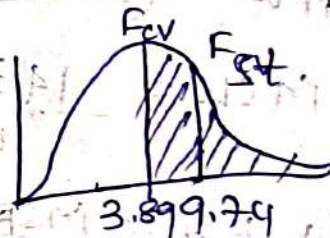
$$MSW = \frac{1860.8}{12} = \boxed{155.1}$$

$$vii) f_{st} = \frac{MSB}{MSW} = \frac{1511.2}{155.1} = \boxed{9.74}$$

$$F_{cv}(0.05, 2, 12) = \boxed{3.89}$$

$\therefore f_{st} > f_{cv} \Rightarrow$ we reject H_0 .

$$\therefore \mu_1 \neq \mu_2 \neq \mu_3 //$$



③ Using 1-way ANOVA perform test at $\alpha = 0.01$ for the given Data.

n	Mean	std. dev
30	50.76	10.45
30	45.32	12.76
50	53.67	11.46

$$\text{Sol.} \rightarrow 1. H_0: \mu_1 = \mu_2 = \mu_3 \quad | \quad H_1: \mu_1 \neq \mu_2 \neq \mu_3$$

$$2. \alpha = 0.01$$

3. Grp means $\Rightarrow 50.76, 45.32, 53.67$

Grand Mean $\rightarrow \bar{x} = \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3}{3} = 49.92$

iv) $SSB \Rightarrow \sum n(x_i - \bar{x})^2$
 $= 30(50.76 - 49.92)^2 + 30(45.32 - 49.92)^2 + 50(53.67 - 49.92)^2$
 $= 1359.093$

~~SSB~~ $df_B = 3 - 1 = 2$

$MSB = \frac{SSB}{df_B} = \frac{1359.093}{2} = 679.55$

v) $SSW \Rightarrow \sum (x_i - \bar{x})^2$
 $= \sum_{i=1}^n (x_i - \bar{x})^2 = (n-1) \cdot s^2$

$SSW = 29(10.45)^2 + 29(12.76)^2 + 49(11.46)^2$
 $= 14323.83$

$df_W = (30+30+50) - 3 = 110 - 3 = 107$

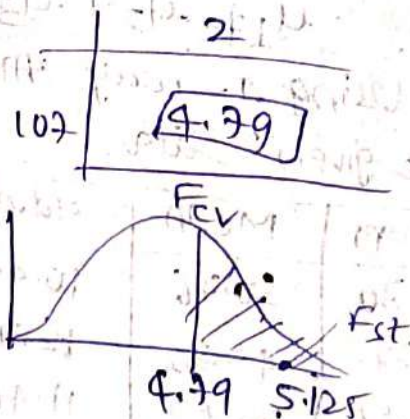
$MSW = \frac{14323.83}{107} = 133.8$

vii) $F_{st} = \frac{MSB}{MSW} = \frac{679.55}{133.8} = 5.12$

$F_{cv}(0.01, 2, 107) = 4.79$

$\therefore F_{st} > F_{cv} \Rightarrow$ we reject Null hypothesis (H_0)

$\therefore \mu_1 \neq \mu_2 \neq \mu_3 //$



(4) An education researcher is comparing 4 diff algebra curriculum. 8th grade students are randomly assigned to 1 of the 4 groups.

Their state achievement scores are compared at the end of the Year. Use the appropriate statistical procedure to determine whether the curriculum differ with respect to Maths achievements. Test at $\alpha=0.05$

n	mean	std
50	170.5	14.5
50	168.3	12.8
50	169.6	17.7
50	172.8	16.8

Sol: i) $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$ | $H_1: \mu_1 \neq \mu_2 \neq \mu_3 \neq \mu_4$

ii) $\alpha = 0.05$

iii) $\bar{x}_1 = 170.5$ $\bar{x}_2 = 168.3$ $\bar{x}_3 = 169.6$ $\bar{x}_4 = 172.8$

$$\bar{x} = \frac{170.5 + 168.3 + 169.6 + 172.8}{4}$$

$$= \boxed{170.3}$$

iv) $SSB = \sum n(x_i - \bar{x})^2$

$$= 50(170.3 - 170.5)^2 + 50(168.3 - 170.3)^2 +$$

$$50(170.3 - 169.6)^2 + 50(170.3 - 172.8)^2$$

$$= \boxed{31476.5} \quad \boxed{539}$$

$df_B = k - 1 = 4 - 1 = \boxed{3}$

$MSB = \frac{539}{3} = \boxed{179.6}$

v) $SSW = 49(14.5)^2 + 49(12.8)^2 + 49(17.7)^2 +$

$$49(16.8)^2$$

$$= \boxed{47511.38}$$

$df_W = 200 - 4 = \boxed{196}$

$MSW = \frac{47511.38}{196} = \boxed{242.405}$

$$vii) f_{st} = \frac{MSB}{MSW} = \frac{179.6}{242.905} = \boxed{0.741}$$

$$vii) F_{cv} (0.05, 3, 196) = \boxed{2.68}$$

$\therefore \boxed{F_{st} < F_{cv}} \Rightarrow$ we do not reject
 H_0

$$\therefore \boxed{\mu_1 = \mu_2 = \mu_3 = \mu_4} //$$

$$\begin{array}{r|l} & 3 \\ \hline 196 & \boxed{2.68} \end{array}$$

